

Self-repairing and anti-fouling performance of anticorrosive coating in marine environment

Xiao Wu^a, Chao Yang^b, Lingli Wu^c, Chuchu Zhang^d, Gan Cui^{d,**}, Yanping Xin^{a,*}

^a Shandong Institute of Petroleum and Chemical Technology, Dongying, 257061, China

^b Technology Inspection Center of Shengli Oilfield, SINOPEC, Dongying, 257000, China

^c Engineering Cost and Cost Control Center, China Petroleum Engineering & Construction Corporation Southwest Company, Chengdu, 610041, China

^d College of Pipeline and Civil Engineering in China University of Petroleum (East China), Qingdao, 266580, China

ARTICLE INFO

Keywords:

Double-walled microcapsules
Fluoropolymer
Anti-corrosion and anti-fouling coating
Marine environment
Electrochemical test

ABSTRACT

Microcrack damage repair and microbial contamination prevention are two major challenges for anti-corrosion coatings on the surfaces of underwater installations. In this study, water-triggered self-healing microcapsules were combined with low-surface-energy fluoropolymers to prepare a bifunctional anti-corrosion coating with underwater self-healing and bio-fouling properties. The results showed that the prepared microcapsules had good mechanical, thermal, and environmental stability, with an average diameter of 153 μm and an encapsulation rate of 66.7%. When the microcapsule content in the coating was 8 wt%, the microcapsule dispersion performance and healing barrier effect of the composite coating were optimal, and the repair efficiency reached 81.6%. Furthermore, compared with the coating without microcapsules or graft fluoropolymers (PFG), the sample denoted as PFG18/4/IPDI-MC/ER showed the largest surface water contact angle and had the best protein inhibition rate of 87% when using bovine hemoglobin to detect the protein adsorption concentration on the coating surface.

1. Introduction

Marine pollution is the accumulation of marine organisms such as bacteria, micro-algae, diatoms, and invertebrates on the surfaces of underwater installations that interfere with the normal operation of the equipment [1,2]. Biofilms are the main cause of marine biological pollution [3].

Anti-fouling coatings have been widely used in ships and other marine equipment as effective means of preventing and controlling biological pollution. Initially, anti-fouling paints were used to stop pollution, mainly through the slow release of fouling agents. However, over time, the disadvantages of using toxic anti-fouling agents have gradually emerged. The introduction of such foreign toxic materials can mutate marine life and damage marine ecosystems [4]. Therefore, it is both urgent and necessary to develop an environmentally friendly anti-fouling coating. Fluoropolymers have excellent properties such as hydrophobicity, weather resistance, chemical resistance, and low surface energy [5,6]. When applied to anti-fouling coatings, fluorine atoms are distributed on the surface of the coating through crosslinking, which

effectively minimizes the rearrangement and osmosis of marine adherents. However, the adhesion of fluoropolymers is poor. In practical applications, special treatment must be used to enhance the bonding force of fluoropolymers with the underlying epoxy resin, which increases the difficulty of operation. The synthesis of block polymers by combining functional monomers is an effective means of overcoming the limitations of fluoropolymer applications. By controlling the number, type, and arrangement of functional monomers, the surface properties of the grafted coating can be changed and modified directionally [7–9].

In addition to bio-fouling, marine anti-corrosion coatings face another problem, namely, microcrack repair in the underwater environment [10]. The underwater environment is complex and changeable, which making the protective coating of the device highly susceptible to microcracks as a result of impact or aging that are difficult to repair. Inspired by the self-healing phenomenon of skin in organisms in nature, researchers have explored various self-healing materials and technologies [11,12], especially for the microcapsule repair method. With cracks appear, the outer wall of the microcapsules ruptures to release internal repair substances, repairing the damaged parts. Unlike conventional

* Corresponding author.

** Corresponding author.

E-mail addresses: chennacuigan@163.com (G. Cui), 2018001@sdipct.edu.cn (Y. Xin).

<https://doi.org/10.1016/j.polymeresting.2023.108090>

Received 3 April 2023; Received in revised form 9 May 2023; Accepted 25 May 2023

Available online 26 May 2023

0142-9418/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

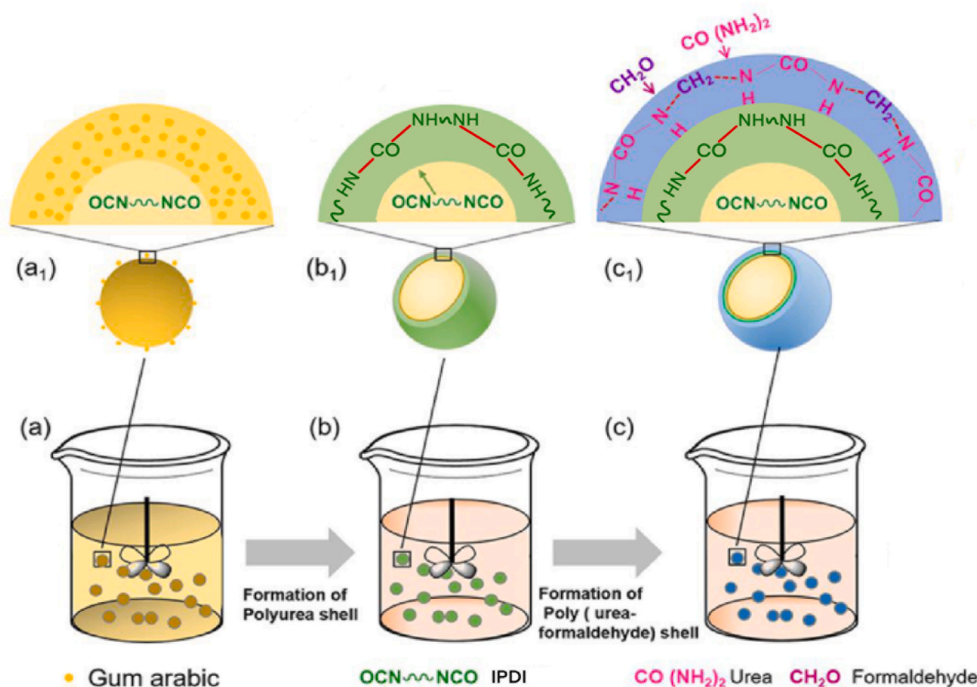


Fig. 1. Preparation process of IPDI-PU/PUF-MCs.

two-component repair methods [13], one-component healing coatings do not require the introduction of catalytic substances, but more common environmental catalysts [14–16]. Among one-component curing agents, water-based isocyanates have attracted widespread attention owing to their simple operation and good anti-corrosion properties [17, 18]. Yang et al. [19] first developed microencapsulation technology for isophorone diisocyanate (IPDI) in 2008. They exploited the difference in reactivity to enable toluene diisocyanate (TDI) prepolymers to form shells at the oil–water interface of the droplets through interfacial polymerization. However, the size of the prepared microcapsules was in the range of 40–400 μm , which was not suitable for application in thin coatings. The authors found that microcapsules approximately 100 μm in size are more in line with the development requirements of epoxy coatings with crack self-healing technology [19].

Furthermore, to improve the stability of microcapsules, researchers have adopted many methods for improving the outer wall of microcapsules, such as introducing inorganic fillers [20], increasing the crosslinking density [21], and constructing a multi-layer structure [22]. Caruso et al. [23] improved the mechanical strength of microcapsules by thickening their shell walls. Sun et al. [24] successfully synthesized a polyurea shell with a bilayer structure and developed diisocyanate (HDI) microcapsules with excellent resistance to organic solvents. In the actual research process, the optimal microcapsule shell modification scheme should be selected by considering the requirements of the actual application environment, properties of the coating matrix, production cost, and complexity of the preparation process. In this study, double-walled microcapsules with IPDI as the core material and polyurea-formaldehyde and polyurethane as the shells were prepared by in situ interfacial hybrid polymerization and embedded in an epoxy matrix. Next, a two-step casting method was used to graft fluoropolymers (PFG) onto the surface of the epoxy resin matrix to obtain bifunctional anti-corrosion coatings (PFG/IPDI-MC/ER). This multi-functional anti-corrosion coating could have great application value in the field of marine protection.

2. Experiment settings

2.1. Materials

Urea, formaldehyde (37% aqueous solution), ammonium chloride, sodium chloride, magnesium sulfate, calcium chloride, magnesium chloride, resorcinol, n-octanol, gum arabic (GA), isophorone diisocyanate (IPDI), ethyl acetate (EA), hydrochloric acid, polyurethane prepolymer (Bayer L-75), and absolute ethanol were purchased from Sinopharm Group in China. Aladdin provided hexafluorobutyl methacrylate, glycidyl methacrylate, azobisisobutyl cyanide, N,N-dimethylformamide, disodium hydrogen phosphate, and sodium dihydrogen phosphate. The epoxy resin primer and curing agent were purchased from Zhenjiang Danbao Resin Co., Ltd. Q235 steel with dimensions of $25 \times 50 \text{ mm}^2$ was purchased from Tianjin Iron and Steel Co., Ltd. Bovine serum albumin (BSA), 98.0% purity, was supplied by Aladdin. All other chemicals were of commercial analytical grade and used as standard.

2.2. Preparation of the microcapsules

Double-walled microcapsules were prepared by in situ interfacial hybrid polymerization. The specific experimental process is shown in Fig. 1. First, GA emulsifier (13.5 g) was added to deionized water (120 mL), and the mixture was mechanically stirred at 500 rpm for 3 h until it was clear and transparent. Then, urea (3.75 g), ammonium chloride (0.38 g), and resorcinol (0.38 g) were added to the emulsion in turn. The pH was adjusted to approximately 3.5 using 0.1 mol/L hydrochloric acid, and a drop of octanol was added to remove surface air bubbles from the mixed solution. A mixture of L-75 (9 g) and EA (9 mL) was stirred for 30 min to obtain a homogeneous prepolymer solution. IPDI (17 g) was then added. The mixture was slowly added dropwise to the GA mixture to form an emulsion. After 10 min, 9.5 g of formaldehyde solution (37%) was gradually added dropwise to the reaction system. Finally, the system was heated to 55 $^{\circ}\text{C}$ for 3 h to complete the reaction. The resulting microcapsule suspension was then cooled to room temperature. The microcapsules were separated using a Buchner funnel, washed several times with deionized water, and dried by blasting for 48 h to obtain the dried microcapsule product. Microcapsules suitable for

Table 1

Monomer and initiator feeding amounts for PFG synthesis.

Mole ratio	Samples	HFBM (g)	GMA (g)	DMF (g)	AIBN (g)
20/2	PFG _{20/2}	0.9096	0.0517	8.6517	0.0095
18/4	PFG _{18/4}	0.8186	0.1034	8.2980	0.0091
14/8	PFG _{14/8}	0.6367	1.2058	7.5915	0.0183

Note: Mole ratio denoted as the mole ratio of HFBM to GMA.

epoxy matrices can be obtained by optimizing the experimental parameters.

2.3. Preparation of fluorinated copolymer

A free-radical polymerization method was used to prepare a random copolymer of hexafluorobutyl methacrylate (HFBM) and glycidyl methacrylate (GMA), denoted as PFGX/Y, where X/Y is the molar ratio of HFBM to GMA. The operation steps were as follows: under nitrogen protection, HFBM and GMA were added to a 50 mL three-necked flask. Anhydrous dimethylformamide (DMF) was then added, and the mixture was fully dissolved under magnetic stirring. Then, the three-necked flask was placed in a constant-temperature water bath with a digital display stirrer. Dry nitrogen gas was bubbled through the flask to remove air from the reaction apparatus. The equipment was heated to 70 °C, and the initiator azobisisobutyronitrile (AIBN) was rapidly added to the reaction device. The mixture was allowed to react for 6 h. After naturally cooling to room temperature, the product was dialyzed three times with DMF and distilled water, and a pure white powder was obtained after freeze-drying. In this experiment, three fluorinated polymers with different monomer molar ratios were obtained. The molar ratios of HFBM to GMA were 20/2, 18/4, and 14/8. The corresponding fluorinated polymers are denoted as PFG_{20/2}, PFG_{18/4}, and PFG_{14/8}. The total number of moles of monomer in this reaction was 0.004 mol, and the amounts of monomer and initiator are shown in Table 1.

2.4. Preparation of PFG/IPDI-MC/ER coating

The preparation process for the PFG/IPDI-MC/ER coating is shown in Fig. 2. First, Q235 steel ($50 \times 25 \text{ mm}^2$) was polished with 400-, 800-, and 1200-mesh sandpaper in sequence. The test pieces were washed with petroleum ether and absolute ethanol and then dried in a vacuum oven. The epoxy resin E44 and polyamide curing agent in a mass ratio of 1:1 were stirred together and mixed evenly. Using absolute ethanol as a solvent, the microcapsules were dispersed in the epoxy resin matrix and vacuum was applied at room temperature for 10 min to remove the remaining air bubbles. The mixture was uniformly coated on the Q235 steel using a wire rod coater and cured at room temperature for 3 h to obtain the uncured self-healing epoxy coating IPDI-MC/ER. The incompletely cured IPDI-MC/ER coating was then dip-coated with a 5 mg/mL PFG solution [the solvent was phosphate buffered saline (PBS), pH = 7.4] and reacted at 55 °C for 2 h. After the reaction, the coating surface was rinsed with a large amount of PBS solution to remove unreacted PFG. Finally, the system was vacuum cured at 25 °C for 4 h to prepare a bifunctional anti-corrosion coating (PFG/IPDI-MC/ER). Epoxy samples and bifunctional anti-corrosion coatings with different molar ratios were prepared using the same procedure and are denoted as ER, PFG_{20/2}/IPDI-MC/ER, PFG_{18/4}/IPDI-MC/ER, and PFG_{14/8}/IPDI-MC/ER.

2.5. Characterization

2.5.1. Characterization of microcapsule properties

The morphology of the microcapsules was observed by field-emission scanning electron microscopy (SEM; S-4800, Hitachi, Japan). The scratched area of the self-healing coating and the dispersion distribution of the microcapsules were observed using a three-dimensional (3D) confocal microscope (Zeiss Axio Imager A2m, Germany). The samples were analyzed for chemical functional groups using Fourier transform-infrared spectroscopy (FT-IR; Thermo Fisher Science Inc.). Thermogravimetric analysis (TGA) was performed using a TGA/DSC1 apparatus (Mettler-Toledo, Switzerland), where the samples were placed in an alumina crucible and heated at a rate of 10 °C/min under

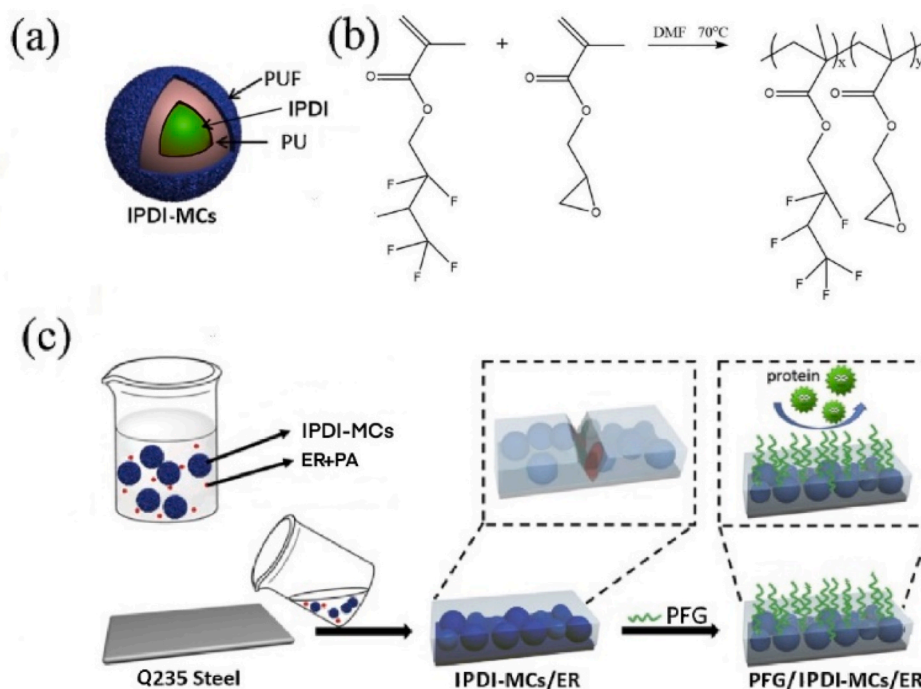


Fig. 2. Preparation process of multifunctional anticorrosive coating: (a) Schematic structure of double-walled microcapsules, (b) Synthesis reaction formula of PFG, and (c) Preparation of composite coating.

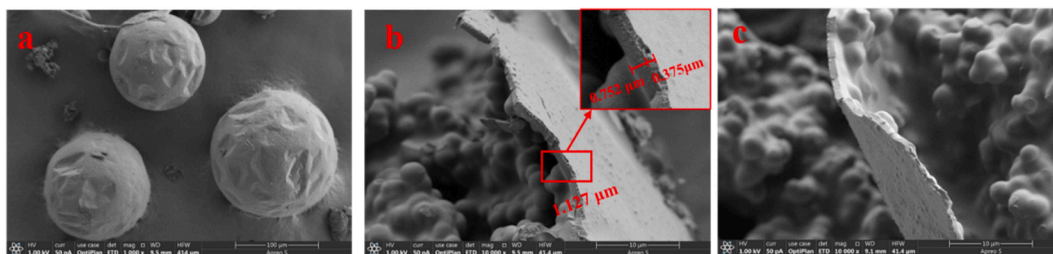


Fig. 3. SEM images of microcapsules: (a) Double-shelled microcapsules, (b) Broken double-shelled microcapsule, and (c) Internal morphology of microcapsules.

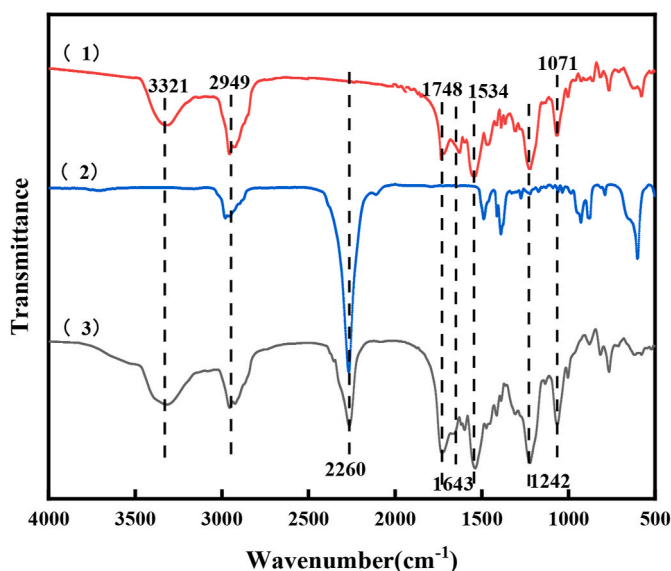


Fig. 4. FTIR spectra of the microcapsules: (1) PUF/PU shell derived from microcapsules, (2) Neat IPDI, and (3) Complete microcapsules.

nitrogen atmosphere. The weight change over a temperature range of 50–500 °C was recorded.

2.5.2. Characterization of self-healing anti-corrosion properties of coatings

X-shaped scratches reaching to the metal substrates were carved on epoxy coatings with different microcapsule contents, which were then immersed in experimental saline. Studies have shown [24] that the reaction curing time of IPDI is 6–8 h. To estimate the release amount of microcapsules more accurately, the experimental soaking time was selected as 8 h. The repair efficiency (β) of the scratches was calculated using Eq. (1):

$$\beta = (H_0 - H_8) / H_0 \times 100\% \quad (1)$$

where, H_0 was the initial depth at 0 h, and H_8 was the depth at 8 h. Due to differences in manual operations, the size of H_0 was subject to the specific experimental object.

Salt spray and electrochemical tests were used to evaluate the anti-corrosion performance of the microcapsule-based self-healing coatings and to optimize the amount of microcapsules added. For the salt spray test, X-shaped scratches were made into the surfaces of the samples. The samples were then exposed to 5 wt% NaCl salt spray. The salt spray box was kept at 35 °C and the pH was 6.5–7.2. The samples were evaluated for their resistance to the salt spray according to the ISO 4628 standard. Electrochemical tests were performed using a PARSTAT 2273 electrochemical workstation. In the three-electrode system, the reference electrode was a saturated calomel electrode (SCE), and the auxiliary electrode was a platinum electrode. The frequency range and sinusoidal voltage of the AC impedance test were 10 kHz to 0.01 Hz and 10 mV,

respectively. The back and sides of the coated test piece were sealed with epoxy resin, and a blade was used to scratch an “X” on the surface of the working electrode. The sample was then connected to the three-electrode system and soaked in 3.5 wt% NaCl solution.

2.5.3. Characterization of protein anti-fouling properties of coatings

The synthetic components and chemical structure of PFG were analyzed using FT-IR and proton nuclear magnetic resonance (^1H NMR). For ^1H NMR, heavy water (D_2O) was used as the solvent, and the frequency was 500 MHz. The resonant hydrogen nuclei positions on the PFG were marked in the spectrum.

PFG/IPDI-MC/ER was successfully prepared using a two-step casting method. The graft density (M) of PFG can be calculated from Eq. (2):

$$M = (m_1 - m_0) / S \quad (2)$$

where, M in mg/cm^2 was the graft density of PFG, m_0 and m_1 were the mass of the coating before and after grafting, respectively, and S was the surface area of the Q235 steel.

To test the protein anti-fouling performance of the sample surface, a BSA protein adsorption experiment was carried out first. Fifty milligrams (50 mg) of BSA was dissolved in PBS (10 mL). The resulting solution was diluted in a 50 mL volumetric flask to obtain a 1 mg/mL BSA/PBS solution, which was used immediately. Subsequently, a sterilized 12-well cell culture plate ($20 \times 20 \text{ mm}^2$) was rinsed three times with PBS, and 2 mL of the BSA/PBS solution was added. The samples pre-soaked in PBS solution for 30 min were removed, dried with nitrogen, and then transferred to the BSA/PBS solution for 24 h. The samples were then rinsed with PBS solution and deionized water three times successively to wash off the adsorbed loose proteins and salts and then dried with nitrogen.

The relative nitrogen content was determined by XPS (X-ray photoelectron spectroscopy) to quantitatively evaluate the amount of protein adsorption on the surface of the PFG/IPDI-MC/ER. In addition, the anti-protein adsorption properties of the composite coatings with different monomer ratios were evaluated, and the optimal monomer molar ratio was determined.

3. Results and discussion

3.1. Preparation of microcapsules

To maintain the high encapsulation efficiency and good spherical morphology of the IPDI-MCs in the epoxy matrix, single-variable optimization experiments were performed to determine the best preparation process parameters. The internal and external appearances of the optimized microcapsules were observed using SEM, and the results are shown in Fig. 3. The double-walled microcapsules were spherical and uniformly dispersed, with an average diameter of 150 μm [Fig. 3(a)]. Fig. 3(b) shows a cross-sectional view of a broken microcapsule shell. The microcapsule shell layer consists of an inner porous PU layer and an outer dense PUF layer. Its thickness was approximately 1.127 μm ; the PU layer was 0.752 μm thick and the PUF layer was 0.375 μm thick [Fig. 3(a)]. As shown in Fig. 3(c), the smooth inner wall of the microcapsules

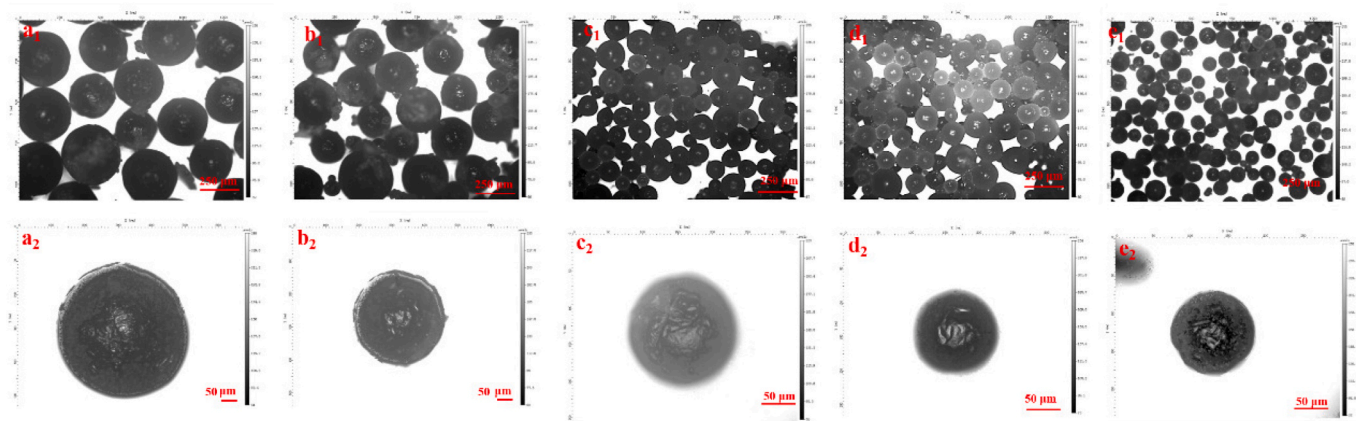


Fig. 5. Images of microcapsules under different agitation rates: (a) 400 r/min, (b) 500 r/min, (c) 600 r/min, (d) 700 r/min, and (e) 800 r/min.

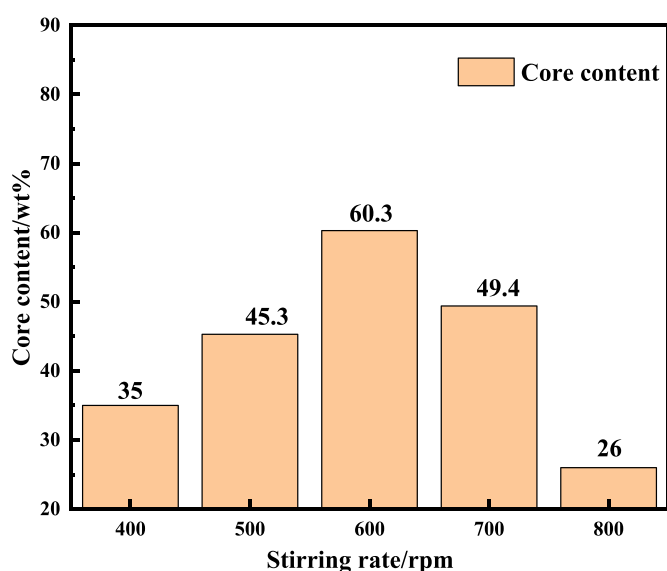


Fig. 6. Effect of different stirring rates on core content.

was favorable for the outflow of the core material, whereas the rough outer wall was favorable for embedding with the epoxy resin. In addition, the polar groups in the PUF layer can form hydrogen bonds with the epoxy resin, which enhances the compatibility of the microcapsules with the epoxy matrix.

The chemical compositions of the microcapsules, shells, and IPDI were characterized by FT-IR, as shown in Fig. 4. The peaks at 2260 cm^{-1} and 3321 cm^{-1} indicated $\text{N}=\text{C}=\text{O}$ vibration and $\text{N}-\text{H}$ stretching vibration, respectively, indicating that IPDI was successfully coated. The vibration of the characteristic $\text{C}=\text{O}$ bond of polyurethane (1748 cm^{-1}) and the tensile vibration of the characteristic $\text{C}=\text{O}$ bond (1643 cm^{-1}), the deformation vibration of $\text{C}-\text{N}$ (1242 cm^{-1}), and the deformation vibration of $\text{N}-\text{H}$ of polyurea-formaldehyde (1534 cm^{-1}) indicated that the microcapsule shell was composed of PU and PUF. The stretching vibration of $\text{C}-\text{H}$ at 2949 cm^{-1} and the peak of $\text{C}-\text{O}-\text{C}$ at 1070 cm^{-1} confirmed the presence of methylene and ether bonds, respectively. The SEM and FT-IR results proved that double-walled core-shell microcapsules with IPDI as the core and PU/PUF as the shell were successfully synthesized.

3.2. Optimization of synthesis parameters of microcapsules

To increase the efficiency and improve the optimization effect, mature parameters such as the type, concentration, reaction

temperature, and pH of the emulsifier were kept unchanged in this study. Parameters such as rotational speed, prepolymer reaction amount, and curing time were optimized [25].

As shown in Fig. 5, microcapsules with regular morphology and uniform dispersion were successfully formed at 400–800 rpm. Manual counting showed that the size distribution of the microcapsules conformed to the normal distribution, and the average sizes at 400, 500, 600, 700, and 800 rpm were 382, 275, 153, 120, and 110 μm , respectively. The reason for this distribution was that as the stirring speed increased, the average shearing force also increased, and thus the size of the droplets (core material) decreased, resulting in smaller microcapsules. In addition, some studies have found that the chemical stability of larger microcapsules was better [26], and when the epoxy matrix was cracked, the repair effect of microcapsules with a diameter of 150 μm was the best [27].

The microcapsule encapsulation rate was measured using the grinding and washing method. First, the dried microcapsules were frozen with liquid nitrogen and pulverized, weighed as w_1 , then fully ground, washed with EA several times by suction filtration until the core material was completely removed, filtered and dried, and weighed as w_2 . The encapsulation rate was calculated using Eq. (3). This core material extraction process was carried out three to five times, and the final results were averaged.

$$\text{Core content}(\%) = (w_1 - w_2) / w_1 \times 100 \quad (3)$$

The encapsulation rates of microcapsules at different stirring speeds are shown in Fig. 6. With increasing stirring speed, the coating rate of microcapsules initially increased and then decreased. The encapsulation rate of the microcapsules reached a maximum of 60.3% at 600 rpm. The reason for this result was that in the coating stage, a low rotational speed resulted in insufficient emulsification, and thus the structural stability of the microcapsules was poor. Furthermore, the particle size of the capsule gradually decreased with the increase of rotational speed as shown in Fig. 5. Moreover, the increase in rotational speed resulted in a thinning of the capsule walls and an increase in the irregularity of the microcapsules' shape. For example, there were definite depressions in the surface of the capsule wall at 800 r/min in Fig. 5(e2). Therefore, it can be inferred that too high of a rotational speed may damage the capsule wall and cause leakage of the core material. In addition, microcapsules that are too small may escape through the filter paper during the suction filtration stage [11]. Thus, 600 rpm was selected as the optimal speed for the preparation of the double-walled microcapsules.

Fig. 7 shows the surface morphology and distribution of microcapsules with different prepolymer contents. As the amount of prepolymer increased, the size of the microcapsules decreased, and the dispersion became poorer. The average diameters of the microcapsules were calculated to be 130, 153, and 110 μm for prepolymer contents of 9, 11,

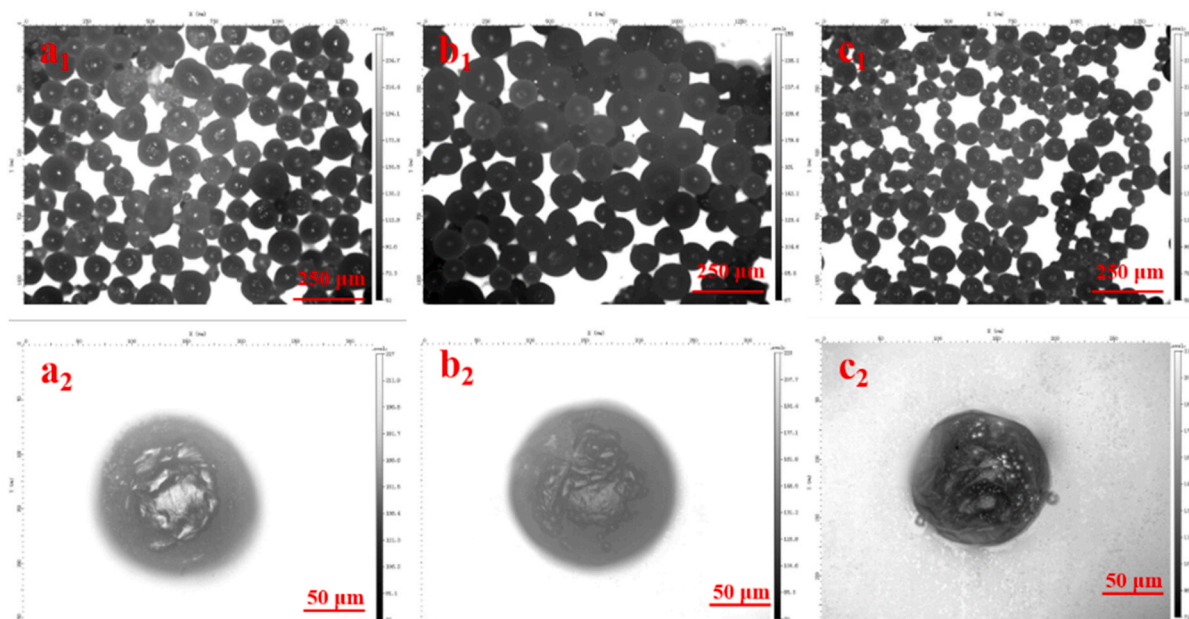


Fig. 7. Images of microcapsules under different prepolymer content: (a) 9 g, (b) 11 g, and (c) 13 g.

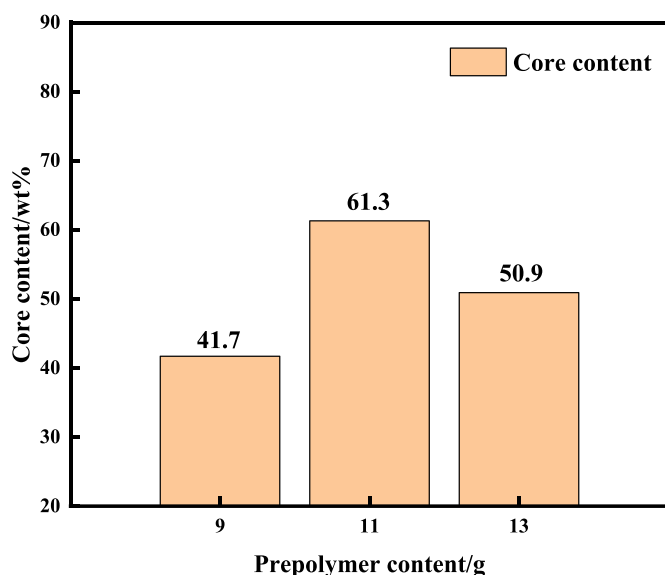


Fig. 8. Effect of different prepolymer content on core content.

and 13 g, respectively. This phenomenon was related to the formation mechanism of the PU shell. If the addition of prepolymer is too small, the prepolymer cannot react in time to form a thin shell attached to the surface of the droplet; the small droplets of the core material collide and develop into large droplets, resulting in the wall being too soft and collapsing. When too much is added, the prepolymer will accumulate on the surface before the droplets collide and aggregate, making the microcapsules smaller. At the same time, too much prepolymer also affects the deposition of the outer layer of PUF particles and deteriorates the dispersion performance of the microcapsules. Through observation of individual microcapsules, it was found that with increasing prepolymer content, the surface wrinkles of the microcapsules deepened, which improved the compatibility between the microcapsules and the epoxy matrix.

The encapsulation rates of microcapsules with different prepolymer contents are shown in Fig. 8. At a prepolymer content of 9 g, the

microcapsules were structurally unstable and easily broken during the drying and filtration process, resulting in an encapsulation rate of only 41.7%. When the prepolymer content was 11 g, the proper amount of prepolymer and shearing speed synergistically helped form stable microcapsules with a proper size. However, when the prepolymer content was increased to 13 g, the microcapsules were small, resulting in a reduced coverage of 50.9%. Therefore, 11 g was selected as the optimal amount of prepolymer.

Fig. 9 shows images of microcapsules under different reaction time. It can be seen that the size of the microcapsules, which was approximately 150 μm with good dispersion, was not affected by the aging time. However, the surface morphology became rougher with aging time because the number of PUF particles deposited on the PU shell increased significantly when the curing time was increased from 150 to 210 min. In addition, it was found that the surface wrinkles of the microcapsules increased with increasing aging time. This was because during the extended curing time, more water molecules diffused into the interior of the microcapsules, reacting with IPDI to generate more CO_2 . The presence of gas created more cavities, resulting in a poor microcapsule structure.

Fig. 10 shows effect of different reaction time on core content. When the curing time was increased from 150 min to 190 min, the encapsulation rate of the microcapsules increased because, with the increased reaction time, the amount of PUF deposition increased, which strengthened the outer wall structure of the microcapsules and reduced the breakage rate of the microcapsules in the subsequent preparation process. When the curing time was increased from 190 to 210 min, the encapsulation rate decreased, the microcapsule structure deteriorated, and the coating rate decreased to 49.4%. Considering the wrinkles and roughness, 190 min was thus selected as the optimal aging time.

3.3. Properties of microcapsules

3.3.1. Thermal stability

Fig. 11 shows the TGA curves of IPDI, PU/PUF, and intact microcapsules. The properties of IPDI (core) and PU/PUF (shell) are relatively stable at 100–200 $^{\circ}\text{C}$ but then they completely decompose at 240 $^{\circ}\text{C}$ and 460 $^{\circ}\text{C}$, respectively. For intact microcapsules, degradation consisted of three stages. In the first stage, from approximately 50 to 200 $^{\circ}\text{C}$, the microcapsules started to lose weight owing to the evaporation of

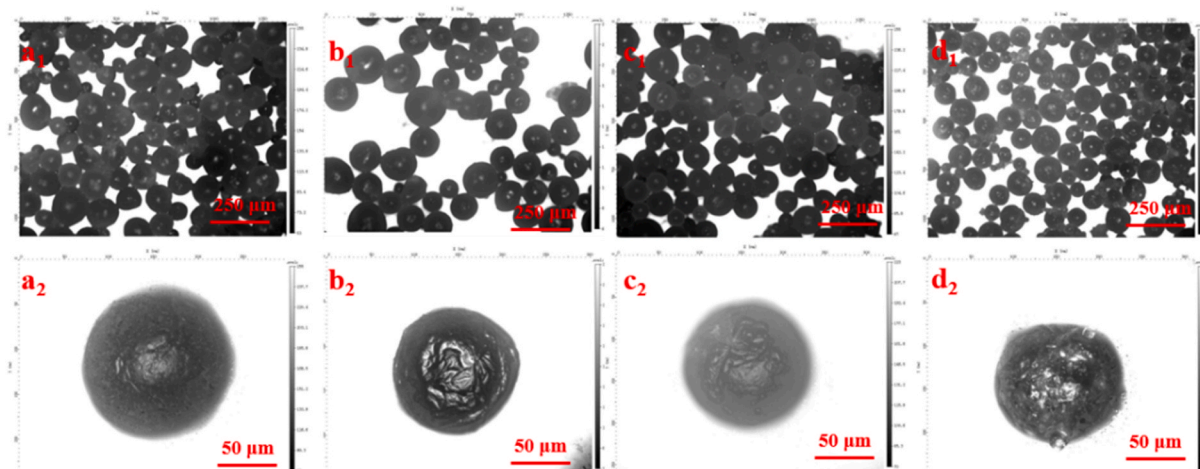


Fig. 9. Images of microcapsules under different reaction time: (a) 150 min, (b) 170 min, (c) 190 min, and (d) 210 min.

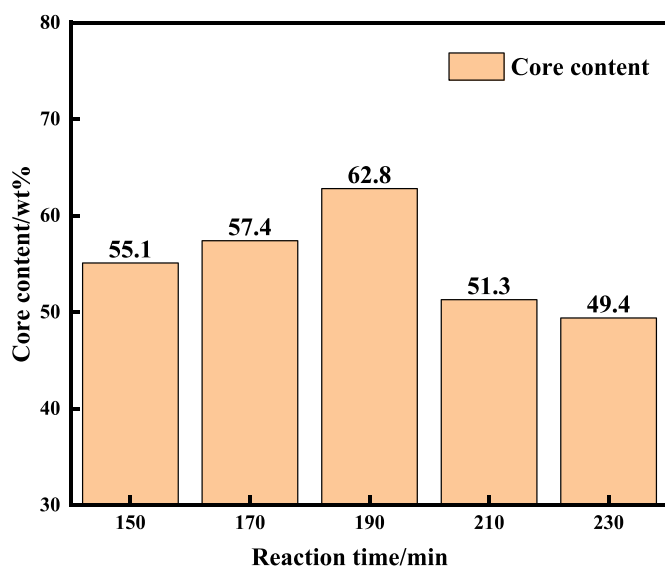


Fig. 10. Effect of different reaction time on core content.

residual solvent (defined as 5% weight loss), but the mass loss in this stage was minimal. In the second stage, the first large-scale weight loss of the microcapsules occurred in an isothermal process at 200 °C (the decomposition temperature of the PU/PUF shell), and the process continued to approximately 340 °C. Compared with IPDI and PU/PUF, the large weight loss of the microcapsules in this stage was mainly affected by the melting of the wall material. This accelerated the destabilization of the microcapsule structure, causing the inner core to lose its protection and partially evaporate. Therefore, the evaporation rate at this stage was substantially higher than that of the outer shell (PU/PUF). Subsequently, the microcapsules began to lose weight for a second time. This process continued up to 460 °C. The weight loss at this stage was mainly owing to further decomposition of the wall material. However, the sustained high temperature also accelerated the flow of polymer chains in the wall material and limited the decomposition of the core material. After decomposition was completed, the remaining specific gravity of the microcapsules was still higher than that of IPDI and PU/PUF. This indicates that the double-walled microcapsules provided protection for the core material.

In general, the microcapsules started to lose a lot of weight after 200 °C. It can thus be considered that 200 °C was the thermal decomposition temperature of the microcapsules and that the microcapsules

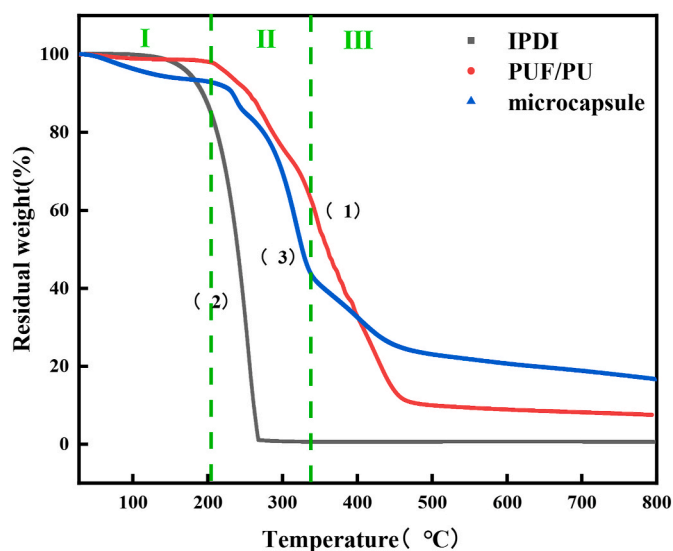


Fig. 11. TGA spectra of the microcapsules: (1) PUF/PU shell derived from microcapsules, (2) neat IPDI, and (3) complete microcapsules.

could be used normally below this temperature. If the processing temperature of the coating matrix is close to the boiling point of the core material or the degradation point of the shell wall, the microcapsule repair function of the core-shell structure will be impaired. Typically, the curing temperature of thermosets such as epoxy is between 100 and 200 °C. This shows that the microcapsules prepared in the experiment can meet most of the coating processing conditions.

3.3.2. Solvent resistance

In this study, a special solution system was constructed to investigate the environmental stability of the microcapsules. Typical organic solvents and artificially simulated seawater were selected as the test solutions, and the environmental stability of the microcapsules was tested by comparing the mass loss and morphology before and after immersion. Simulated seawater was prepared by dissolving NaCl (13.37 g), MgCl₂ (1.13 g), MgSO₄ (1.63 g), and CaCl₂ (0.56 g) in 500 mL of deionized water.

In keeping with studies on microcapsules with similar structures [28, 29], 20 days was chosen as the soaking time in this experiment. The results are shown in Fig. 12. Overall, the microcapsules maintained a complete core-shell structure before and after soaking. However, after

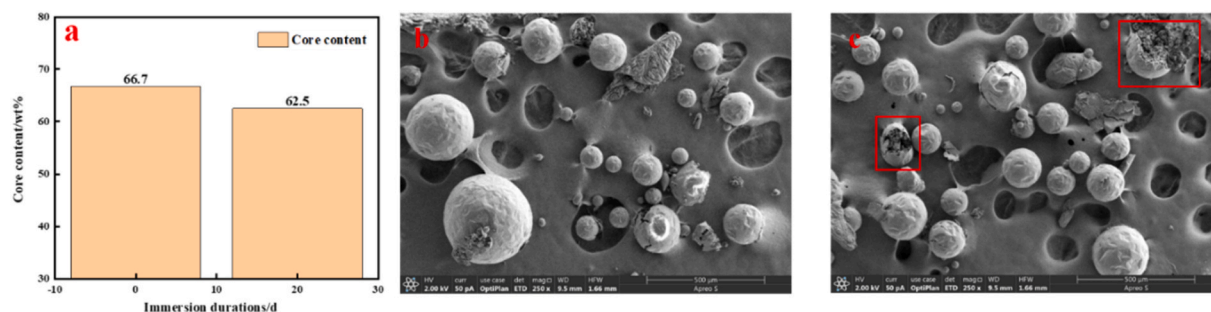


Fig. 12. The immersion experiment of microcapsules in experimental seawater: (a) the change of the core content of the microcapsules before and after immersion, (b) original morphology of microcapsules, and (c) the changing morphology of the microcapsules after immersing 20 days.

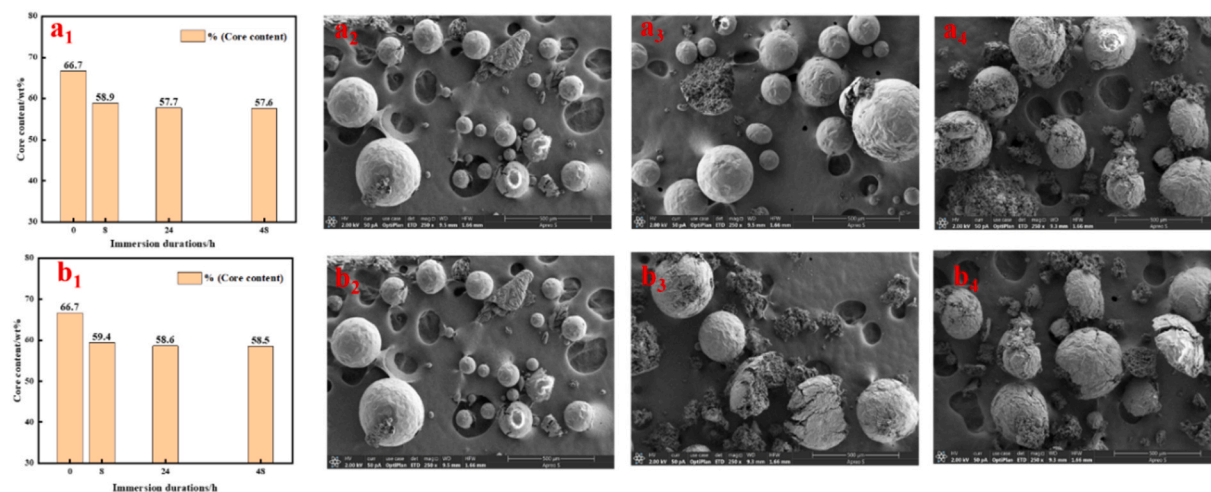


Fig. 13. The immersion experiment of microcapsules in organic solvents: (a) and (b) was anhydrous ethanol and ethyl acetate respectively, (2)–(4) was the immersion time of 0, 24 h and 48 h respectively.

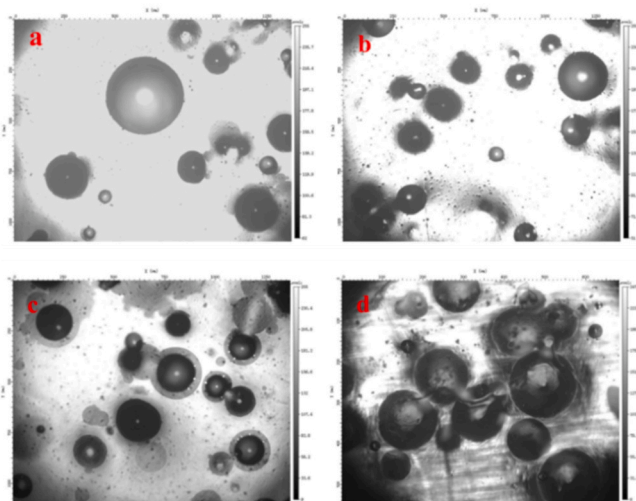


Fig. 14. The dispersion of microcapsules with different mass fractions in the coating matrix: (a) 3 wt%, (b) 5 wt%, (c) 8 wt%, and (d) 10 wt%.

20 days, some microcapsule core materials leaked (marked in the red box) owing to the structural instability of the incompletely cured microcapsules. After soaking for 20 days, the encapsulation rate of the microcapsules decreased from 66.7% to 62.5%, but the overall change did not exceed 5%, and significant consumption of the core did not occur. Compared with previous research results [28,29], it was found

that the microcapsules in this study showed better structural stability performance under longer soaking times.

Considering the preparation time (8 h) and the swelling effect of the organic solvent [26], the tested soaking times of the microcapsules were 8 h, 24 h and 48 h, as shown in Fig. 13. In this paper, ethyl acetate and anhydrous ethanol were chosen for the experiment. After soaking for 24 h, it can be seen that the microcapsules swelled, but the structure remained relatively intact. This is because thermosetting matrices such as polyurethane and polyurea-formaldehyde can swell in organic solvents, and the swelling degree increases with increased soaking time [24]. After soaking for 48 h, most of the microcapsule shells were damaged, and even the core material leaked. The degree of structural damage of microcapsules in absolute ethanol (strong polarity) was higher than that in ethyl acetate (moderate polarity). This is because the swelling ratio of the polymer increased as the polarity of the solvent increased, which accelerated the structural failure of the microcapsules.

The mass loss of microcapsules in ethanol was greater than that in ethyl acetate, but the difference was not large. This is because the hydroxyl groups in ethanol can react with IPDI, consuming the core material to generate new polyurethane. The hydroxyl groups accumulated on the outside of the microcapsules, compensating for the loss of quality caused by core leakage. In the first 8 h, the mass loss rate of the microcapsules was below 8 wt%. Compared with previous research results [24], the results show that microcapsules with a PU-PUF double-shell structure have good solvent resistance.

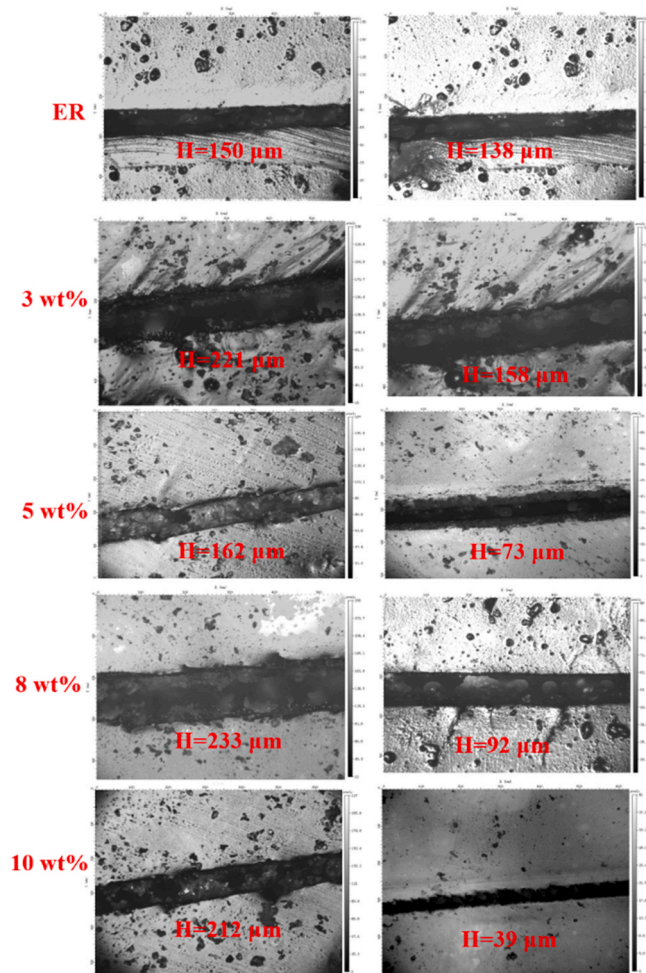


Fig. 15. Scratch repair of the self-healing coating before and after soaking (a) ER; (b) 3-IPDI-MCs/ER; (c) 5-IPDI-MCs/ER; (d) 8-IPDI-MCs/ER; (e) 10-IPDI-MCs/ER.

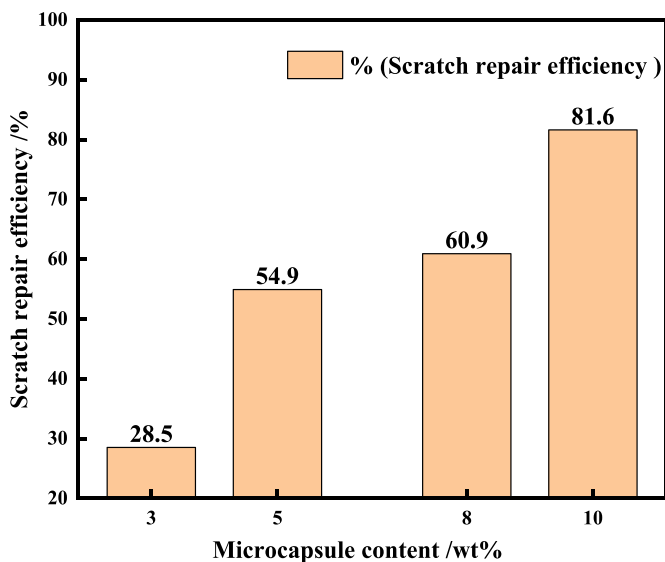


Fig. 16. Scratch healing efficiency of composite coatings with different microcapsule contents.

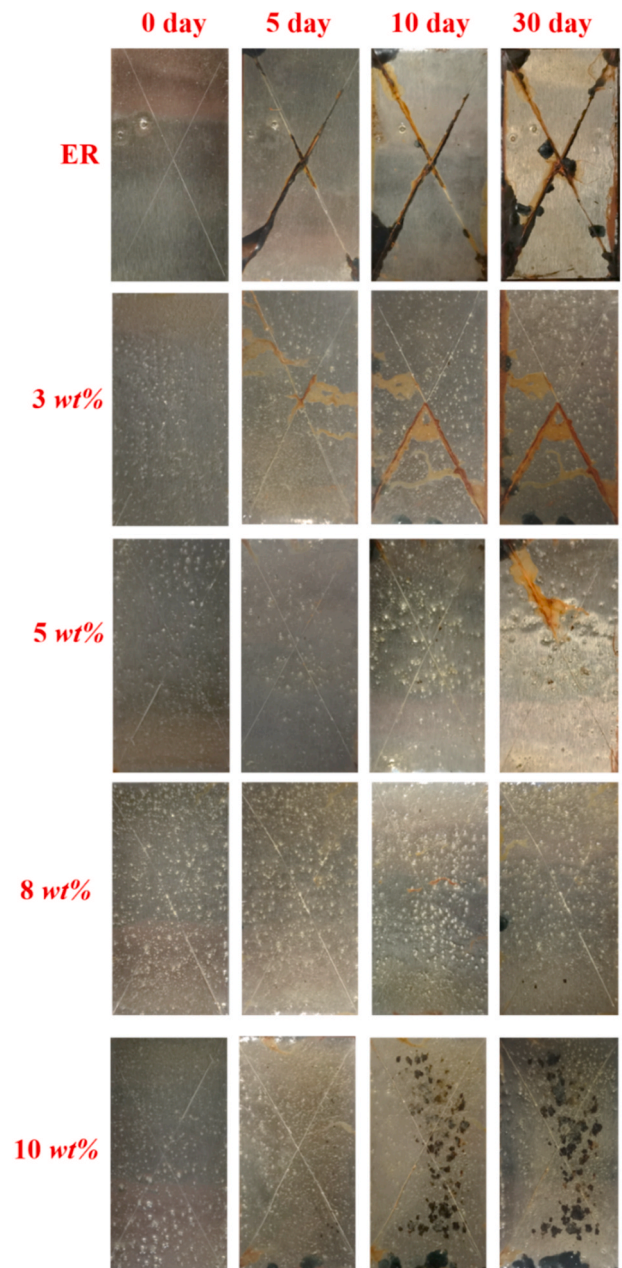


Fig. 17. Morphology of water-triggered self-healing coatings after salt spray test.

3.4. Self-healing anti-corrosion property of IPDI-MC/ER coating

3.4.1. Dispersion of IPDI-MCs in the coating

The content and distribution of microcapsules affect the triggered effect of the self-healing system. The dispersion of the microcapsules was therefore evaluated by adding 3, 5, 8, and 10 wt% microcapsules to the epoxy matrix and observing their morphology in the transparent matrix, as shown in Fig. 14. The composite coatings with 3, 5, and 8 wt% microcapsules had smooth surfaces. However, when the content increased to 10 wt%, it can be seen that the surface where the microcapsules gather is wrinkled. This degraded the mechanical properties of the coating, causing stress concentrations and accelerating coating degradation.

3.4.2. Self-healing performance of IPDI-MC/ER

To further explore the effect of microcapsule content on underwater

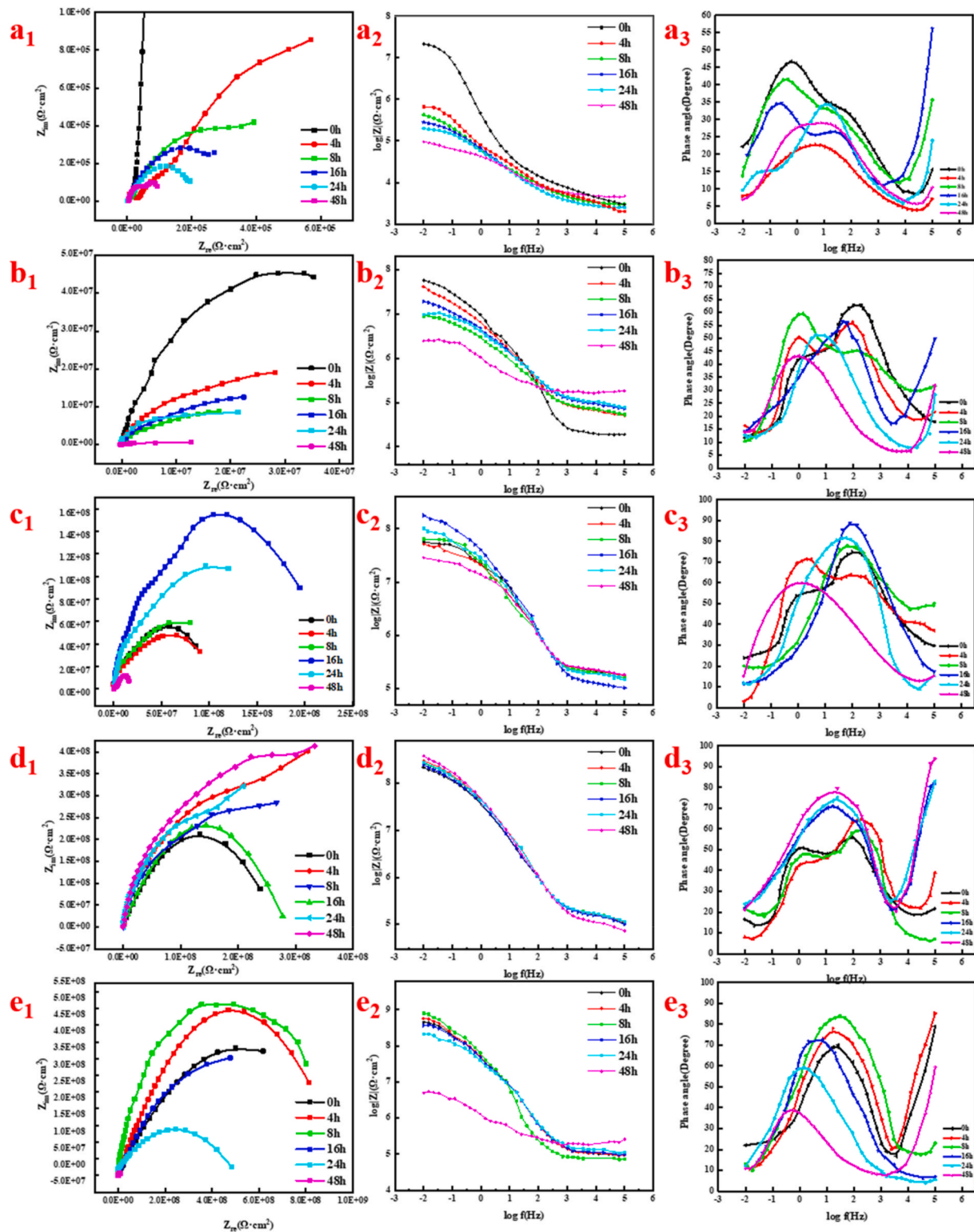


Fig. 18. Nyquist and Bode plots of self-healing coatings with different contents of microcapsules (0, 3 wt%, 5 wt%, 8 wt%, 10 wt%) after soaking for 0, 4, 8, 12, 18, 24 and 48 h.

scratch repair, X-shaped scratches were made into coatings with different microcapsule contents, which were then immersed in experimental seawater. Fig. 15 shows the height distributions of the pure epoxy coatings and self-healing coatings containing 3, 5, 8, and 10 wt% microcapsules before and after repair. In the initial state, the coating matrix is segmented, and the metal substrate is completely exposed to the seawater solution. After soaking for 8 h, the scratch width decreases

owing to the inherent elasticity of the epoxy matrix. The restoration of the pure epoxy coating is shown in Fig. 15. It can be observed that the depth of the scratch changed by 12 μm after soaking for 8 h. The degree of change was small, presumably because of the corrosion of metals in the salt-containing solution, resulting in the accumulation of some corrosion products. This also shows that the crack condition did not improve after the pure epoxy coating matrix was damaged. Compared

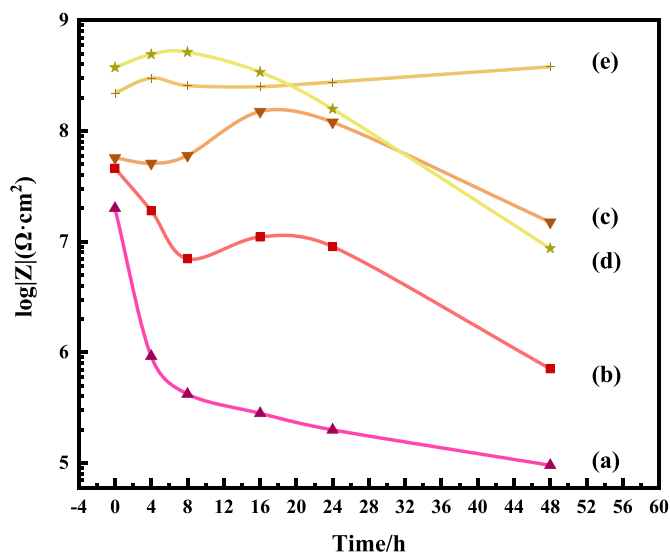


Fig. 19. Low-frequency impedance values of self-healing coatings at different immersion times: (a) ER, (b) 3-IPDI-MCs/ER, (c) 5-IPDI-MCs/ER, (d) 8-IPDI-MCs/ER, and (e) 10-IPDI-MCs/ER.

with the pure epoxy matrix, the scratch depth of the composite coating matrix with the addition of microcapsules was significantly reduced. As the microcapsule content increased, the difference in scratch depth decreased. At 10 wt% microcapsules, the repaired scratch depth was only 39 μm .

The results of Eq. (1) show that the repair efficiencies of the composite coatings with 3, 5, 8, and 10 wt% microcapsule content were 28.5, 54.9, 60.5, and 81.6%, respectively (Fig. 16). This is because when scratches occurred, the peripheral microcapsules ruptured and released the healing agent (IPDI) under capillary action. IPDI reacted with water and cured into polyurethane, which sealed the scratched position. With the increase in the amount of microcapsules, the healing substances released under the triggering of scratches increased until the scratches were completely filled and level with the original substrate surface.

3.4.3. Anti-corrosion performance of IPDI-MC/ER

Fig. 17 shows the corrosion morphologies of the self-healing coatings with different microcapsule contents in the salt spray test at 0, 5, 10, and 20 days. For the pure epoxy on Day 5, there are visible signs of corrosion in the scratched area and the accumulation of corrosion products. Over time, corrosion developed beneath the coating and spread. At 30 days, the coating lifted at the scratch and “bubbling” occurred. Compared with the epoxy coating, the corrosion of the composite coating obviously improved, and the size of the corrosion area decreased with increasing microcapsule content. When the microcapsule content was 8 wt%, the yellow corrosion products and black corrosion areas disappeared. There was no obvious corrosion of the coating throughout the experimental period, with the exception of edge corrosion caused by improper sealing of the edge of the test piece, which is not considered here. However, when the microcapsule content was 10 wt%, the covered substrate surface began to generate randomly distributed black spots in the non-scratched area on the 10th day, and the size of the area increased significantly with time. Combined with the results presented in Section 3.4.1, it can be seen that the microcapsules aggregated with each other owing to excessive addition. This aggregation weakened the bonding effect between the coating and matrix and accelerated the infiltration of corrosive media at the edge. In addition, the closed coating led to the continuous accumulation of corrosive media, resulting in the multi-point corrosion phenomenon in the non-scratched area. Combined with the scratch test results, it can be concluded that the composite coating with 8 wt% microcapsules had the best anti-corrosion effect.

Fig. 18 shows the Nyquist and Bode plots of the self-healing coatings with different contents of microcapsules after soaking for 0, 4, 8, 12, 18, 24, and 48 h. The anti-corrosion performance of coatings is usually evaluated using the low-frequency impedance modulus, $|Z|_{0.01\text{Hz}}$, as shown in Fig. 19. It can be seen that the $|Z|_{0.01\text{Hz}}$ of the epoxy resin decreases continuously from the initial 20 $\text{M}\Omega\text{cm}^2$ to 0.0952 $\text{M}\Omega\text{cm}^2$. This indicates that the shielding performance of the broken coating without microcapsules decreased with increasing penetration of the corrosive media. In addition, the $|Z|_{0.01\text{Hz}}$ of the self-healing coatings were all higher than those of the epoxy coatings. This indicates that the addition of microcapsules enhanced the physical shielding ability and anti-corrosion performance of the epoxy matrix. Observing the $|Z|_{0.01\text{Hz}}$ at the beginning of immersion, it was found that with the increase in the amount of microcapsules, the $|Z|_{0.01\text{Hz}}$ increased. This shows that as the microcapsule content increased, the initial physical shielding performance of the coating improved. When the microcapsule content was low (3 and 5 wt%), the $|Z|_{0.01\text{Hz}}$ of the composite coatings decreased with increased soaking time in the first 8 h, and the lower the microcapsule content, the greater the rate of decline. This indicates that the corrosive medium began to penetrate the scratched area during the initial immersion, impairing the anti-corrosion performance of the coating. But at 8–16 h, the $|Z|_{0.01\text{Hz}}$ of the coating began to recover. This shows that the scratches started to be repaired after 8 h, which is consistent with the IPDI curing time (8 h) mentioned above. In addition, the repaired polyurethane enhanced the corrosion resistance of the epoxy matrix itself. After 16 h, the low-frequency impedance modulus of the self-healing coating continued to decrease. This shows that the repair substances released by the microcapsules were gradually consumed, and the corrosive medium gradually penetrated the interior of the coating under long-term immersion, resulting in failure of the coating.

When the microcapsule content was high (8 and 10 wt%), the impedance value initially increased. This is because the probability of the microcapsules being triggered and the physical barrier properties of the coating both increased with increasing microcapsule content. Simultaneously, the tertiary amine in the epoxy matrix catalyzed the reaction between IPDI and free hydroxyl groups to form hydrogen bonds, which accelerated the repair process of the coating. However, after 8 h, the barrier properties of 10-IPDI-MC/ER began to decline. Combined with the results of the salt spray test, it was found that at 10 wt% microcapsules, too many microcapsules agglomerated. The microcapsules in the initial stage acted only as a barrier medium, enhancing the physical shielding ability of the coating. However, with increasing penetration time, the agglomeration of the microcapsules deteriorated the bonding effect between the coating and the metal substrate, resulting in the gradual failure of the anti-corrosion performance of the self-healing coating. In conclusion, when the microcapsule content was 8 wt%, the water-triggered self-healing coating (8-IPDI-MC/ER) exhibited the best anti-corrosion performance.

To further study the repair-failure process of the coating, a long-cycle electrochemical test was conducted. Fig. 20 presents the long-cycle electrochemical test results for 8-IPDI-MC/ER. The phase-angle diagram of the initial self-healing coating contains three time constants, which are attributed to the interfacial reaction of the metal substrate, the interstitial layer formed by the contact between the two sides of the scratch, and the coating matrix. There were two time constants in the middle and late stages of the experiment. The intermediate-frequency time constant is derived from the resistance and capacitance inside the composite coating matrix, whereas the low-frequency time constant is related to the electrochemical corrosion reaction at the coating/substrate interface. With the penetration of water and electrolyte ions, the coating resistance decreased, and the frequency phase angle gradually decreased from 90°.

To distinguish the action stages of the coating, the $|Z|_{0.01\text{Hz}}$ at different time points were extracted, and the numerical changes are shown in Fig. 21. The repair-failure process of the coating is divided into four stages [17], namely, pre-self-healing (I), self-healing (II), moisture

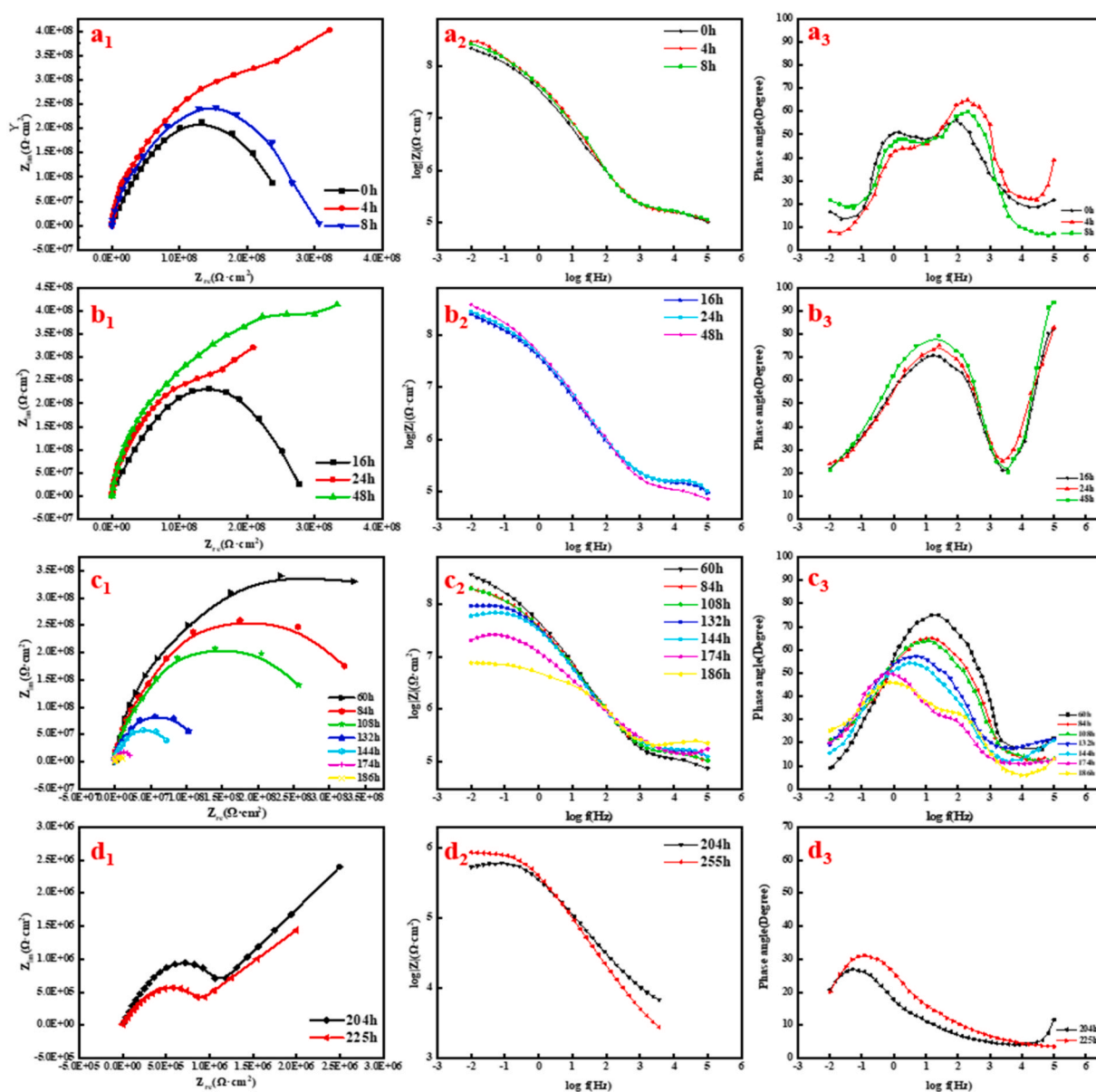


Fig. 20. Nyquist and Bode plots of 8-IPDI-MCs/ER at different repair stages: (a) Nyquist plot, (b) Amplitude-frequency characteristic curve of Bode plot, and (c) Phase-frequency characteristic curve of Bode plot.

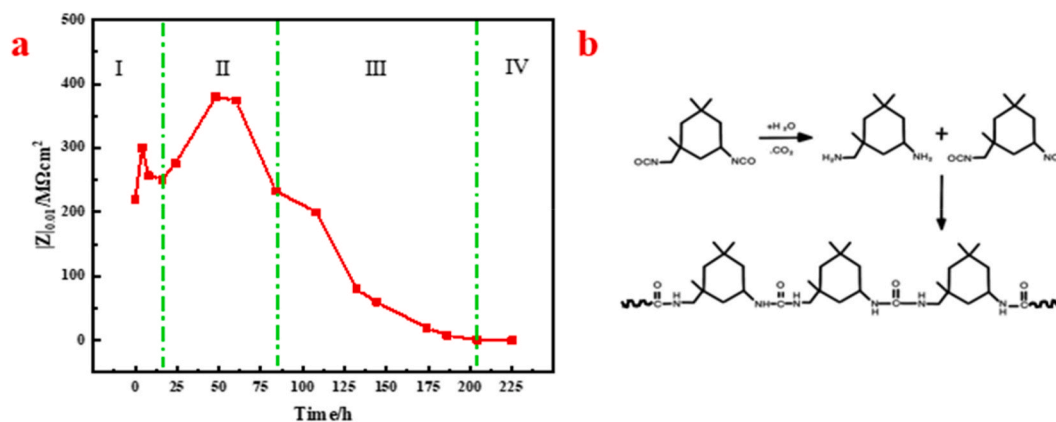


Fig. 21. Time-varying low frequency impedance modulus of 8-IPDI-MCs/ER (a) and IPDI's water-triggered curing reaction mechanism (b).

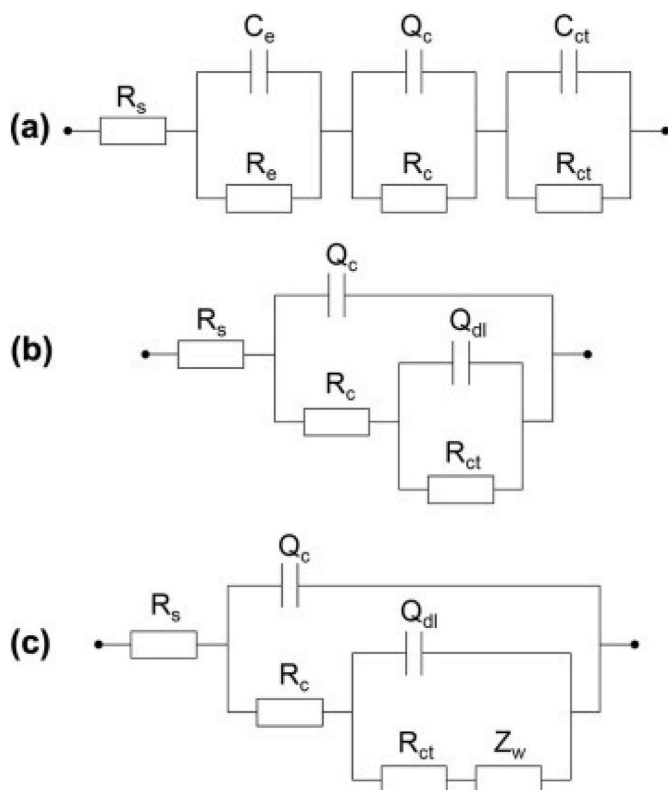


Fig. 22. Equivalent circuit of scratched 8-IPDI-MCs/ER in corrosive medium.

intrusion (III), and coating failure (IV). During the first phase, $|Z|_{0.01\text{Hz}}$ increased slightly because the two sides of the scratch were in contact with each other under the inherent elasticity of the material, which hindered the penetration of the solution in the crack. During the self-healing stage (16–84 h), the corrosion solution began to penetrate the scratches and reacted with IPDI to form insoluble polyurethane. As the immersion time increased, an increasing number of cracks were repaired, and the coating impedance reached a peak value ($380 \text{ M}\Omega \text{ cm}^2$). However, when IPDI was consumed (84 h), $|Z|_{0.01\text{Hz}}$ declined under continuous penetration of the corrosive medium. $|Z|_{0.01\text{Hz}}$ showed a fluctuating downward trend (from 380 to $0.932 \text{ M}\Omega \text{ cm}^2$) in Phase 3, which indicates that moisture was beginning to invade. However, the infiltration of water was blocked by the microcapsules in the matrix, and the microcapsules swelled under long-term soaking to enhance the

barrier effect. This intermittent infiltration of water was repeated until the substrate was completely penetrated and $|Z|_{0.01\text{Hz}}$ decreased below 10^6 . At this point, the coating completely lost its ability to resist corrosive media.

Three equivalent circuits were constructed for the corrosion stages described above (see Fig. 22). Equivalent circuit (a) was chosen to simulate the corrosion state of the coating in the pre-self-healing stage (I). Equivalent circuit (b) was chosen to fit the repair process of the coating during the self-healing (II) and moisture intrusion stages (III). Finally, the Warburg impedance was introduced for equivalent circuit (c) to fit the corrosion medium diffusion phenomenon at the metal interface during the coating failure stage (IV).

In the equivalent circuits, R_s represents the resistance of the corrosion solution; R_e and C_e are the resistance and capacitance of the coating itself, respectively; R_c and Q_c are the resistance and capacitance of the coating with scratches, respectively; R_{ct} and Q_{dl} are the charge transfer resistance and electric double-layer capacitance at the metal interface, respectively; and Z_w is the Warburg diffusion coefficient.

The fitting results of the equivalent circuit are shown in Fig. 23. As the immersion time increases, R_c exhibits roughly the same trend as the low-frequency impedance, whereas Q_c has the opposite trend. This is consistent with the electrochemical information represented by the

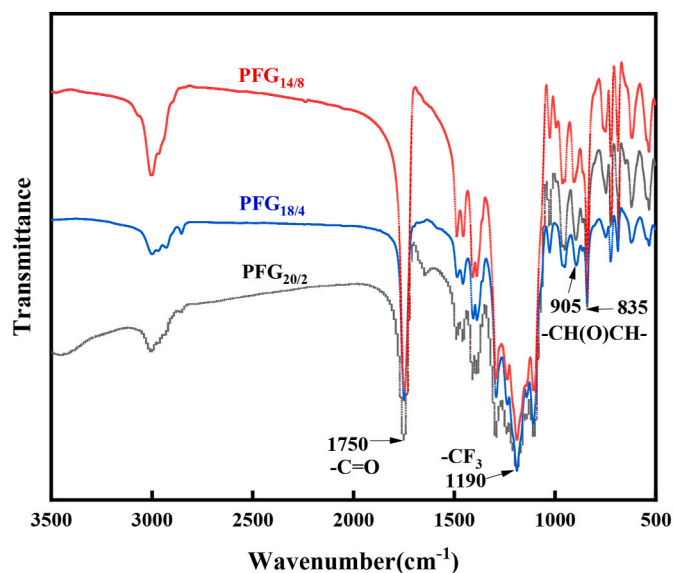


Fig. 24. FTIR spectrum of PFG.

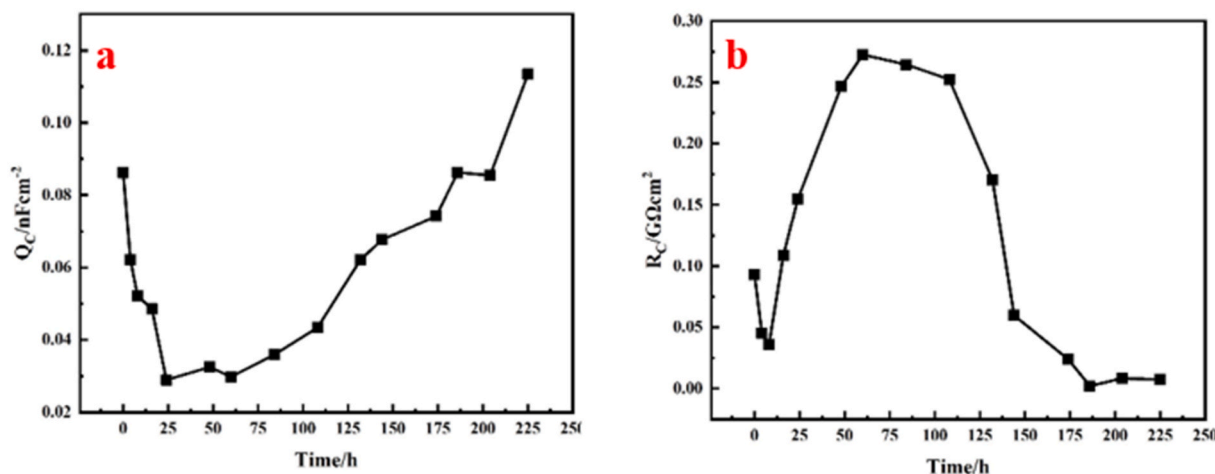


Fig. 23. Fitting results of equivalent parameters of 8-IPDI-MCs/ER with scratches in corrosive medium: (a) Q_c and (b) R_c .

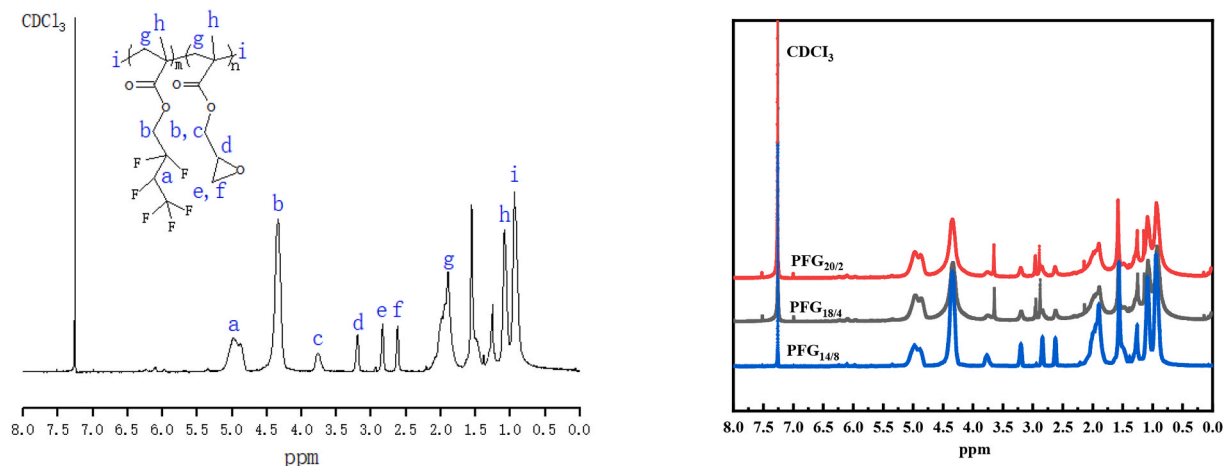
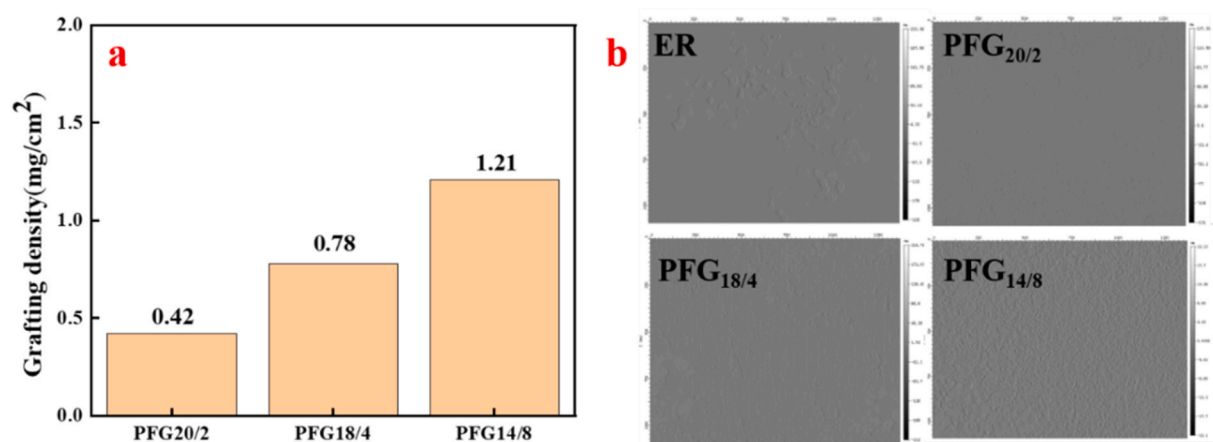
Fig. 25. ^1H NMR spectrum of PFG.

Fig. 26. The grafting density of PFG in different molar ratios (a) and surface topography of epoxy coating and multifunctional anticorrosive coating (b).

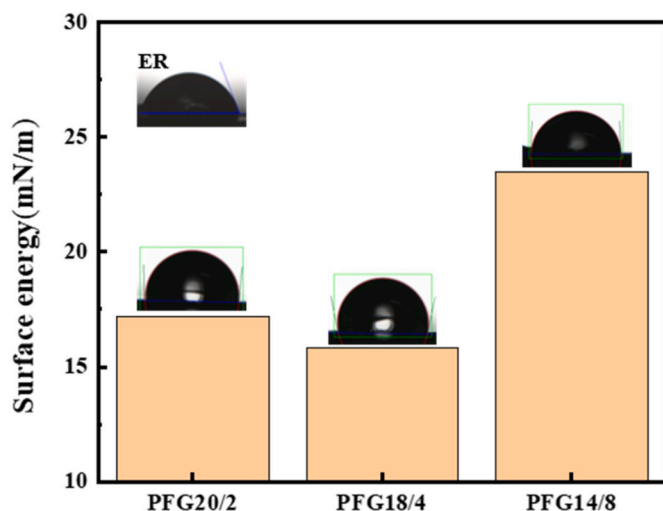


Fig. 27. Surface energy and water contact angle of multifunctional anticorrosive coatings with different monomer molar ratios.

equivalent parameters. As the penetration resistance of the coating increased, R_c increased and Q_c decreased.

3.5. Synthesis of PFG

In this study, fluorine-containing HFBM and epoxy group-containing GMA were used as reaction monomers, and azobisisobutyl cyanide was used as the initiator to synthesize PFG using the traditional free-radical polymerization method. The FT-IR spectra (Fig. 24) reflect the characteristic groups and proportions in the polymer. The stretching vibrations of the methyl group and methylene group are in the range of $3000\text{--}3040\text{ cm}^{-1}$, the stretching vibration peak of the CF_3 group is located at 1180 cm^{-1} , the characteristic peaks at 1750 cm^{-1} are from the $\text{C}=\text{O}$ stretching vibration of the ester group, and the characteristic absorption peaks of the epoxy groups are located at 905 cm^{-1} and 840 cm^{-1} . With increasing GMA content, the peak of the $-\text{CH}(\text{O})\text{CH}-$ group increases. This preliminarily proves that fluorinated polymers with different ratios were successfully synthesized.

In the ^1H NMR characterization of the polymer (Fig. 25), the multiplets at $1.0\text{--}1.5\text{ ppm}$ indicate the absorption peaks of hydrogen atoms (h) and trace amounts of hydrogen atoms (i) connected to the PFG main chain. The absorption peak at 1.82 ppm indicates methylene hydrogen atoms (g). The peaks at 2.55 ppm and 2.72 ppm are the absorption peaks of the epoxy group methylene (e, f), and the absorption peak at 3.32 ppm represents the epoxy group methine (d). The single peaks at 3.75 , 4.32 , and 5.0 ppm are the absorption peaks at c, b, and a, respectively, on the copolymer chain. With increasing GMA content, the absorption peaks of

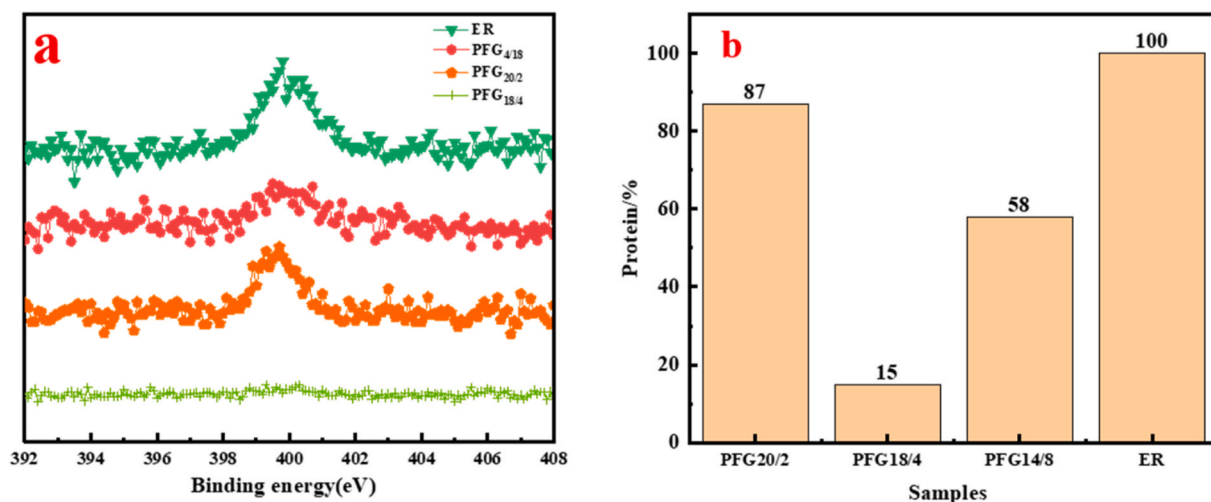


Fig. 28. N1s (a) and adsorption equivalent (b) of the multifunctional anticorrosive coating on the XPS spectrum after BSA protein adsorption.

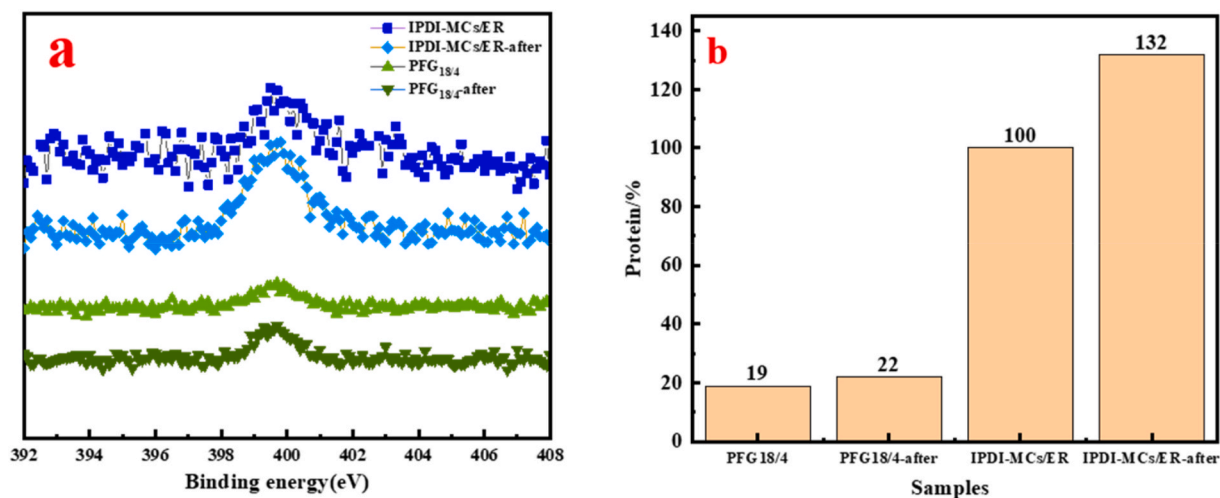


Fig. 29. Antifouling performance of the multifunctional coating before and after damage repair: N_{1s} image (a) and adsorption equivalent (b) of the coating surface after BSA protein adsorption.

methylene (e, f) and methine (d) on the epoxy groups increased. Combined with the FT-IR results, these results indicate that PFG with different molar ratios was successfully prepared.

3.6. Properties of the PFG/IPDI-MC/ER coating

3.6.1. Surface morphology of the PFG/IPDI-MC/ER coating

The anti-fouling performance of PFG/IPDI-MC/ER mainly depends on the ratio of the low-surface-energy components and the surface roughness of the coating [30,31]. The surface morphology of the composite coating is shown in Fig. 26. It can be seen that the surface of the composite coating without PFG is covered with circular depressions of different sizes, whereas the surface depressions disappear into granular protrusions after grafting PFG. This indicates that PFG was successfully grafted. Before grafting, the evaporation of the solvent and the addition of microcapsules resulted in the formation of pits on the surface of IPDI-MC/ER. The incompatibility and solubility differences of the different components in the PFG caused the copolymers to exhibit micron-to nanometer-size phase regions. These microphase-separated structures were the reason for the granular protrusions on the surface after grafting.

In addition, it can be seen that the roughness of PFG/IPDI-MC/ER increases with increasing GMA content. On the one hand, the increase

in GMA content introduces more epoxy groups, increasing the number of reaction sites, which improves the connection effect between PFG and the epoxy matrix. The interaction between the components (GMA and HFBM) was also enhanced with increasing GMA content. The increased degree of microphase separation on the surface also increased the surface roughness of the composite coating. On the other hand, the decrease in HFBM represents a decrease in the number of hydrophobic fluorine atoms. According to the Gibbs free energy formula, fluorine-containing segments with low surface energies tend to enrich the surface under non-polar conditions. Therefore, the selection of the monomer molar ratio should comprehensively consider the above two aspects and use the results of the water contact angle to determine the optimal molar ratio.

3.6.2. Static water contact angle of PFG/IPDI-MC/ER coating

The water contact angles of the composite coatings are shown in Fig. 27. IPDI-MC/ER is hydrophilic, with a contact angle of only 75°. The contact angles of the coatings improved after grafting PFG to 108° for PFG_{20/2}/IPDI-MC/ER, 114° for PFG_{18/4}/IPDI-MC/ER, and 84° for (PFG_{14/8})/IPDI-MC/ER. Thus, PFG_{18/4}/IPDI-MC/ER exhibited the largest water contact angle because as the proportion of HFBM decreased, the number of low-surface-energy components decreased. At the same time, the roughness increased with increasing GMA ratio. The difference in the anti-fouling ability of the actual surface mainly lies in

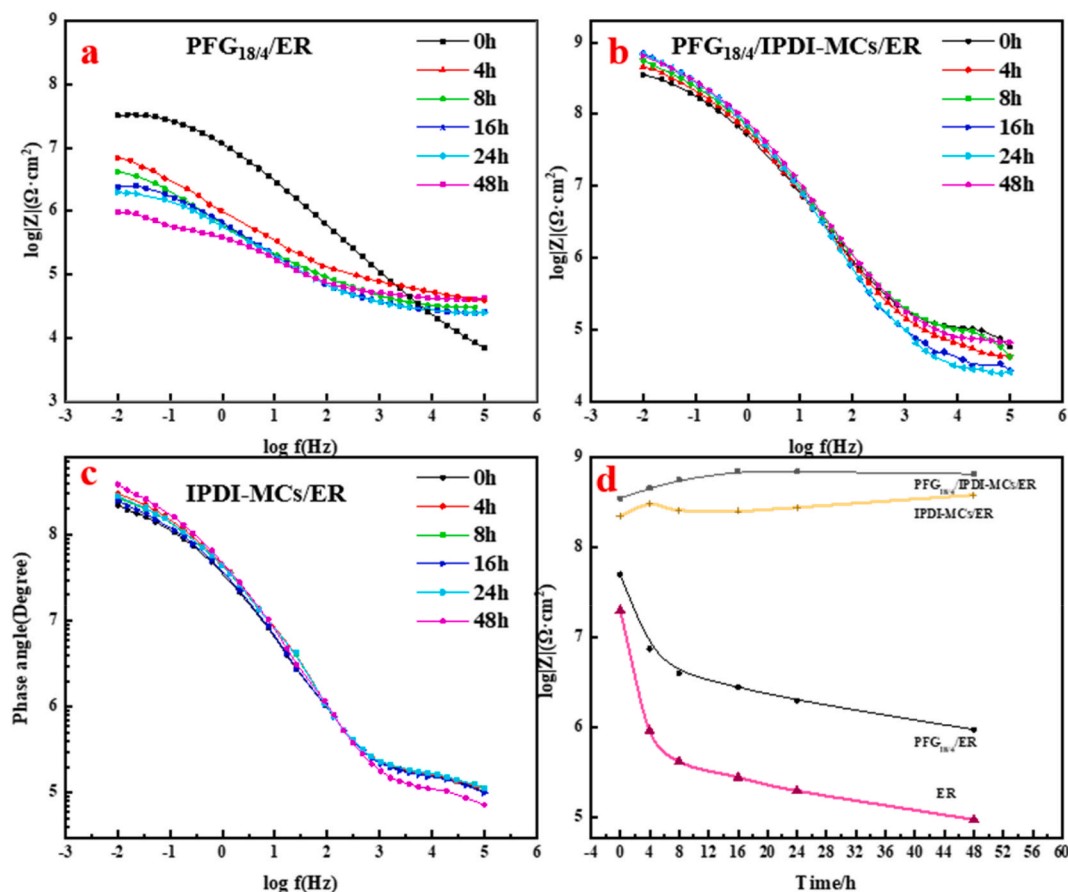


Fig. 30. Bode plot and low frequency impedance plot of damaged multifunctional coating after soaking for 0, 4, 8, 12, 18, 24 and 48 h.

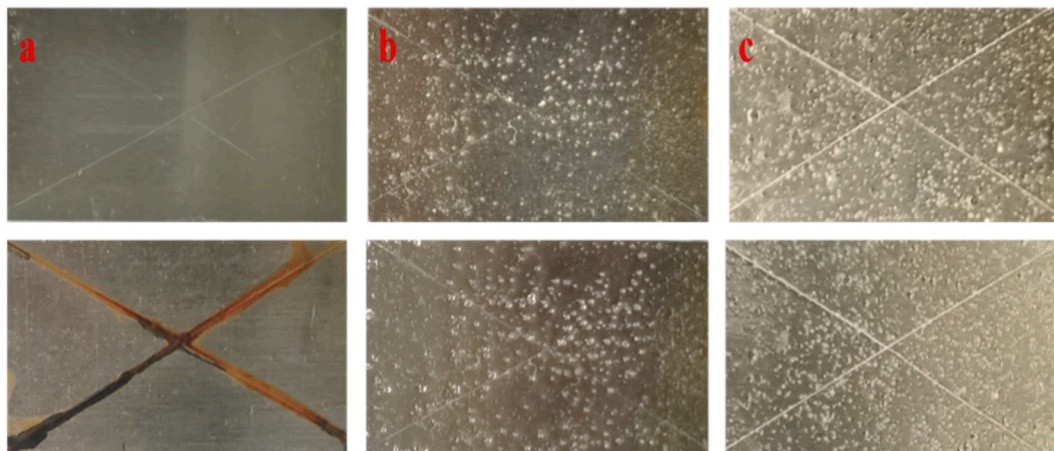


Fig. 31. Self-repair and antifouling effects of the multifunctional anticorrosive coating surface before and after immersion for 7 days: (a) PFG_{18/4}/ER, (b) IPDI-MCs/ER, (c) PFG_{18/4}/IPDI-MCs/ER.

its surface roughness and surface component free energy; a proper monomer ratio helps balance the relationship between the two. Based on the above factors, the water contact angle of the composite coating was the largest when the monomer molar ratio was 18:4.

3.6.3. Protein anti-fouling property of PFG/IPDI-MC/ER coating

To verify the optimal molar ratio of PFG, the actual anti-fouling effect of the composite coating was tested through protein anti-fouling experiments. When the test sample does not contain nitrogen atoms, XPS is a simple and effective method for evaluating the effect of protein

adsorption. Fig. 28 shows an N1s map of the composite coating surface. Assuming that the BSA adsorption capacity of IPDI-MC/ER was 100%, the adsorption capacity obtained by converting the N1s peak integration of each sample into a percentage is shown in Fig. 28(b). The results show that the difference in the monomer ratio can affect the anti-protein properties of the composite coating. Thus, PFG_{18/4}/IPDI-MC/ER exhibited the best protein anti-fouling performance. This proves that the “synergistic effect” can only be exhibited when both the hydrophobic segment and roughness reach a certain value, resulting in excellent anti-protein adsorption performance.

3.7. Multifunctional synergistic properties of PFG/IPDI-MC/ER coating

The multifunctional synergistic effect of the anti-corrosion coatings was explored through self-healing, anti-corrosion, and anti-fouling performance tests. The tests were conducted in two steps. In the first part, PFG18/4/IPDI-MC/ER was used as the control group and IPDI-MC/ER was used as the experimental group. XPS was used to explore whether the multifunctional coating exhibited good bio-fouling performance after damage repair. In the second part, PFG18/4/IPDI-MC/ER was used as the control group, and the composite coatings without microcapsules (PFG18/4/ER) and without PFG (IPDI-MC/ER) were used as the experimental groups. Electrochemical tests and immersion experiments were conducted to explore whether the composite coating after grafting PFG exhibited self-healing and anti-corrosion properties.

In the first part of the experiment, "X"-shaped scratches were made on the surfaces of the samples, which were then immersed in the experimental seawater for 48 h. Subsequently, BSA protein adsorption experiments were performed on the surfaces under the assumption that the BSA adsorption capacity of IPDI-MC/ER before repair was 100%. The test results are shown in Fig. 29. It can be seen that IPDI-MC/ER has more BSA adsorption than PFG18/4/IPDI-MC/ER, which indicates that the anti-fouling performance of the coating after grafting with PFG was improved. In addition, after soaking for 48 h, the protein adsorption of IPDI-MC/ER increased by 32%, indicating that a large amount of BSA was adsorbed onto the surface, and thus the anti-fouling performance was lost. The protein adsorption of PFG18/4/IPDI-MC/ER increased by only 3% after soaking because the hydrophilicity of the epoxy matrix on IPDI-MC/ER facilitated BSA adsorption in solution. At the same time, the scratches accelerated the penetration of water. After soaking for 48 h, the metal substrate under the scratches corroded, and the hydrophobicity of carbon steel decreased, which promoted the further adsorption of BSA. For PFG18/4/IPDI-MC/ER, the hydrophobic structure on the surface of the composite coating was less damaged owing to the elastic shrinkage of the epoxy matrix, and it still had good protein anti-fouling performance. After soaking for 48 h, the scratches were filled with hydrophobic polyurethane formed by IPDI reaction curing. This resulted in protein adsorption of only 22%. Based on the above results, it can be considered that PFG18/4/IPDI-MC/ER damaged by scratches still had good anti-fouling performance.

In the second part of the experiment, Bode and low-frequency impedance change plots of the samples after immersion for 0, 4, 8, 12, 18, 24, and 48 h (Fig. 30). It was found from the low-frequency impedance change diagram [Fig. 30(d)] that the $|Z|_{0.01\text{Hz}}$ of the coating increased after PFG grafting compared with the epoxy resin. This shows that the shielding performance of PFG18/4/ER was improved, and its failure process was slower. Thus, the effective protection period of the coating was increased. The self-healing coating grafted with PFG (PFG18/4/IPDI-MC/ER) also exhibited better shielding performance than the pure self-healing coating (IPDI-MC/ER). However, there were some differences in the repair-failure process between the two. Unlike PFG18/4/IPDI-MC/ER, IPDI-MC/ER did not exhibit initial failure behavior. This is because the presence of PFG and the microphase-separated structure changed the surface properties of the coating, so that the hydrophobic properties of the coating were improved, thereby improving the shielding performance of the coating.

The scratched samples were immersed in simulated seawater for seven days to analyze the effect of PFG on the corrosion resistance of the damaged coatings (the repair process of 8-IPDI-MC/ER lasted for 84 h). Fig. 31 shows the change in the coating morphology before and after immersion. It can be seen that compared with PFG18/4/ER, PFG18/4/IPDI-MC/ER has minimal or no corrosion traces after immersion, which is roughly the same as the results for IPDI-MC/ER. Combined with the electrochemical tests, it can be concluded that the double-walled microcapsule-based epoxy coating after grafting PFG still had good self-healing and anti-corrosion properties. Therefore, it can be considered that the prepared multifunctional coating still had good self-

healing, anti-fouling, and anti-corrosion performance after damage repair.

4. Conclusion

The surfaces of the microcapsules prepared in this study were rough and dense, with an average diameter of 150 μm and an encapsulation rate of 66.7%. The microcapsules exhibited not only good thermal stability (thermal decomposition temperature of 200 $^{\circ}\text{C}$) but also good structural stability performance when immersed in simulated seawater and organic solvents. The salt spray test results showed that with increasing repair time and microcapsule content, the release of the core material increased; the efficiency of scratch repair also increased up to 81.6%. In addition, according to the change in the low-frequency impedance modulus, the healing process of the water-triggered self-healing coating could be divided into four stages: pre-self-healing, self-healing, water permeation, and coating failure. Microscopy and water contact angle test results showed that with increasing GMA content, the graft density and surface roughness between the polymer and the epoxy matrix increased. The results of the protein adsorption experiments showed that the anti-protein adsorption performance of PFG18/4/IPDI-MC/ER was the best, and the amount of protein adsorption decreased to 15%. In addition, through the anti-fouling (XPS characterization) and self-healing anti-corrosion performance tests (electrochemical test and soaking test), it was proven that PFG18/4/IPDI-MC/ER still had good bio-fouling and self-healing anti-corrosion functions after break repair. This effective combination of underwater self-healing and anti-fouling functions provides a new idea for the design of marine anti-corrosion coatings and has good application prospects in the field of underwater industrial facility protection.

Author statement

Xiao Wu: Methodology, Investigation, Experiment, Writing - Original Draft; Chao Yang: Data processing, Writing - Original Draft; Lingli Wu: Investigation; Supervision; Chuchu Zhang: Writing - Review & Editing, Gan Cui: Funding Acquisition, Resources, Supervision, Yanping Xin: Investigation, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no conflicts of interest to report regarding the present study.

Data availability

No data was used for the research described in the article.

Acknowledgements

This work was supported by the Science Development Funding Program of Dongying of China (Grant No. DJ2021006; Grant No. DJ2021008).

References

- [1] L.D. Chambers, J.A. Wharton, R.J.K. Wood, F.C. Walsh, K.R. Stokes, Techniques for the measurement of natural product incorporation into an antifouling coating, *Prog. Org. Coating* 77 (2014) 473–484, <https://doi.org/10.1016/j.porgcoat.2013.11.013>.
- [2] M.S. Selim, M.A. Shenashen, S.A. El-Safty, S.A. Higazy, M.M. Selim, H. Isago, A. Elmarakbi, Recent progress in marine foul-release polymeric nanocomposite coatings, *Prog. Mater. Sci.* 87 (2017) 1–32, <https://doi.org/10.1016/j.pmatsci.2017.02.001>.
- [3] L. Chen, Y. Duan, M. Cui, R. Huang, R. Su, W. Qi, Z. He, Biomimetic surface coatings for marine antifouling: natural antifoulants, synthetic polymers and surface microtopography, *Sci. Total Environ.* 766 (2021), 144469, <https://doi.org/10.1016/j.scitotenv.2020.144469>.

- [4] I. Omae, Organotin antifouling paints and their alternatives, *Appl. Organomet. Chem.* 17 (2003) 81–105, <https://doi.org/10.1002/aoc.396>.
- [5] B.F. Li, B. Qi, Z.Y. Guo, D.X. Wang, Tifeng Jiao, Recent developments in the application of membrane separation technology and its challenges in oil-water separation: a review, *Chemosphere* 327 (2023), 138528, <https://doi.org/10.1016/j.chemosphere.2023.138528>.
- [6] L. Chen, L. Li, X. Zhao, S. Wang, Y. Li, G. Li, X. Ge, L. Wang, Construction of antifouling fluorinated polymer brush via activators regenerated by electron transfer ATRP and thiol-epoxy click reaction, *React. Funct. Polym.* 165 (2021), 104974, <https://doi.org/10.1016/j.reactfunctpolym.2021.104974>.
- [7] Q. Wang, G. Sun, Q. Tong, W. Yang, W. Hao, Fluorine-free superhydrophobic coatings from polydimethylsiloxane for sustainable chemical engineering: preparation methods and applications, *Chem. Eng. J.* 426 (2021), 130829, <https://doi.org/10.1016/j.cej.2021.130829>.
- [8] K. Ravi, W.L. Sulen, C. Bernard, Y. Ichikawa, K. Ogawa, Fabrication of micro-/nano-structured super-hydrophobic fluorinated polymer coatings by cold-spray, *Surf. Coating. Technol.* 373 (2019) 17–24, <https://doi.org/10.1016/j.surfcoat.2019.05.078>.
- [9] Y. Liu, K. Fu, J. Liu, Y. Tian, H. Zhang, R. Wang, B. Zhang, H. Zhang, F. Zhou, Q. Zhang, Design and preparation of a multi-fluorination organic superhydrophobic coating with high mechanical robustness and icing delay ability, *Appl. Surf. Sci.* 497 (2019), 143663, <https://doi.org/10.1016/j.apsusc.2019.143663>.
- [10] B.J. Blaiszik, S.L.B. Kramer, S.C. Olugebefola, J.S. Moore, N.R. Sottos, S.R. White, Self-healing polymers and composites, *Annu. Rev. Mater. Res.* 40 (2010) 179–211, <https://doi.org/10.1146/annurev-matsci-070909-104532>.
- [11] M. Huang, J. Yang, Facile microencapsulation of HDI for self-healing anticorrosion coatings, *J. Mater. Chem.* 21 (2011), 11123, <https://doi.org/10.1039/c1jm10794a>.
- [12] G. Cui, C. Zhang, A. Wang, X. Zhou, X. Xing, J. Liu, Z. Li, Q. Chen, Q. Lu, Research progress on self-healing polymer/graphene anticorrosion coatings, *Prog. Org. Coating* 155 (2021), 106231, <https://doi.org/10.1016/j.porgcoat.2021.106231>.
- [13] Y.C. Yuan, M.Z. Rong, M.Q. Zhang, G.C. Yang, Study of factors related to performance improvement of self-healing epoxy based on dual encapsulated healant, *Polymer* 50 (2009) 5771–5781, <https://doi.org/10.1016/j.polymer.2009.10.019>.
- [14] D. Zhao, M. Wang, Q. Wu, X. Zhou, X. Ge, Microencapsulation of UV-curable self-healing agent for smart anticorrosive coating, *Chin. J. Chem. Phys.* 27 (2014) 607–615, <https://doi.org/10.1063/1674-0068/27/05/607-615>.
- [15] S.J. García, H.R. Fischer, P.A. White, J. Mardel, Y. González-García, J.M.C. Mol, A. E. Hughes, Self-healing anticorrosive organic coating based on an encapsulated water reactive silyl ester: synthesis and proof of concept, *Prog. Org. Coating* 70 (2011) 142–149, <https://doi.org/10.1016/j.porgcoat.2010.11.021>.
- [16] T. Szabó, J. Telegdi, L. Nyikos, Linseed oil-filled microcapsules containing drier and corrosion inhibitor – their effects on self-healing capability of paints, *Prog. Org. Coating* 84 (2015) 136–142, <https://doi.org/10.1016/j.porgcoat.2015.02.020>.
- [17] W. Wang, L. Xu, X. Li, Y. Yang, E. An, Self-healing properties of protective coatings containing isophorone diisocyanate microcapsules on carbon steel surfaces, *Corrosion Sci.* 80 (2014) 528–535, <https://doi.org/10.1016/j.corsci.2013.11.050>.
- [18] E.N. Brown, M.R. Kessler, N.R. Sottos, S.R. White, In situ poly(urea-formaldehyde) microencapsulation of dicyclopentadiene, *J. Microencapsul.* 20 (2003) 719–730, <https://doi.org/10.1080/0265204031000154160>.
- [19] J. Yang, M.W. Keller, J.S. Moore, S.R. White, N.R. Sottos, Microencapsulation of isocyanates for self-healing polymers, *Macromolecules* 41 (2008) 9650–9655, <https://doi.org/10.1021/ma801718v>.
- [20] M.W. Patchan, B.W. Fuller, L.M. Baird, P.K. Gong, E.C. Walter, B.J. Vidmar, I. Kyei, Z. Xia, J.J. Benkoski, Robust composite-shell microcapsules via pickering emulsification, *ACS Appl. Mater. Interfaces* 7 (2015) 7315–7323, <https://doi.org/10.1021/acsami.5b00494>.
- [21] G. Wu, J. An, X.-Z. Tang, Y. Xiang, J. Yang, A versatile approach towards multifunctional robust microcapsules with tunable, restorable, and solvent-proof superhydrophobicity for self-healing and self-cleaning coatings, *Adv. Funct. Mater.* 24 (2014) 6751–6761, <https://doi.org/10.1002/adfm.201401473>.
- [22] S. Kang, M. Baginska, S.R. White, N.R. Sottos, Core-shell polymeric microcapsules with superior thermal and solvent stability, *ACS Appl. Mater. Interfaces* 7 (2015) 10952–10956, <https://doi.org/10.1021/acsami.5b02169>.
- [23] Y. Yang, M.W. Urban, Self-healing polymeric materials, *Chem. Soc. Rev.* 42 (2013) 7446, <https://doi.org/10.1039/c3cs60109a>.
- [24] D. Sun, J. An, G. Wu, J. Yang, Double-layered reactive microcapsules with excellent thermal and non-polar solvent resistance for self-healing coatings, *J. Mater. Chem. A* 3 (2015) 4435–4444, <https://doi.org/10.1039/C4TA05339G>.
- [25] Y.-B. Chong, H. Zhang, C.Y. Yue, J. Yang, Fabrication and release behavior of microcapsules with double-layered shell containing clove oil for antibacterial applications, *ACS Appl. Mater. Interfaces* 10 (2018) 15532–15541, <https://doi.org/10.1021/acsami.8b05467>.
- [26] D. Sun, Y.B. Chong, K. Chen, J. Yang, Chemically and thermally stable isocyanate microcapsules having good self-healing and self-lubricating performances, *Chem. Eng. J.* 346 (2018) 289–297, <https://doi.org/10.1016/j.cej.2018.04.046>.
- [27] S.A. McDonald, S.B. Coban, N.R. Sottos, P.J. Withers, Tracking capsule activation and crack healing in a microcapsule-based self-healing polymer, *Sci. Rep.* 9 (2019), 17773, <https://doi.org/10.1038/s41598-019-54242-7>.
- [28] C. Li, J. Tan, J. Gu, L. Qiao, B. Zhang, Q. Zhang, Rapid and efficient synthesis of isocyanate microcapsules via thiol-ene photopolymerization in Pickering emulsion and its application in self-healing coating, *Compos. Sci. Technol.* 123 (2016) 250–258, <https://doi.org/10.1016/j.compscitech.2016.01.001>.
- [29] H. Yi, Y. Yang, X. Gu, J. Huang, C. Wang, Multilayer composite microcapsules synthesized by Pickering emulsion templates and their application in self-healing coating, *J. Mater. Chem. A* 3 (2015) 13749–13757, <https://doi.org/10.1039/C5TA02288F>.
- [30] B. Zhu, Z. Liu, J. Liu, Y. Yang, Y. Meng, F. Yu, L. Jiang, G. Wei, Z. Zhang, Preparation of fluorinated/silanized polyacrylates amphiphilic polymers and their anticorrosion and antifouling performance, *Prog. Org. Coating* 140 (2020), 105510, <https://doi.org/10.1016/j.porgcoat.2019.105510>.
- [31] R.F. Brady, I.L. Singer, Mechanical factors favoring release from fouling release coatings, *Biofouling* 15 (2000) 73–81, <https://doi.org/10.1080/08927010009386299>.